Adaptive LLM-Based Chatbot with Personality-Aware Dialogue for Human-Centered Applications: A Formal Preliminary Study

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Abstract

Emotional wellbeing and personalized support represent fundamental challenges across diverse human-centered applications, from mental health assistance and educational guidance to customer service and workplace coaching. While conversational AI offers scalable solutions for emotional support and personalized interaction, many existing digital assistants remain generic and fail to adapt to users' individual personality differences. This limits their effectiveness in providing contextually appropriate assistance. This master's thesis specifies a modular, workflow-orchestrated architecture for personality-aware dialogue. We integrate Big Five (OCEAN) personality detection with Zurich Model-aligned behavior regulation to create a reusable research infrastructure supporting domain-specific applications.

Building on Devdas (2025), who demonstrated substantial performance gains using a detection→regulation→generation pipeline, this work extends that foundation by developing a production-ready, reusable system architecture. We optimize for temporal stability, reproducibility, and cross-domain deployment. The system integrates a workflow orchestration backend (N8N visual automation platform) with a RESTful frontend API, enabling clean separation between conversation logic and user interface delivery.

Three core architectural enhancements advance the approach:

1. **Temporal smoothing** with exponential moving average (EMA—a weighted averaging technique giving more weight to recent observations) and confidence-aware filtering stabilizes personality assessment across conversation turns while acting only on reliable trait readings.

2. **Configurable workflow design** with domain-specific policy packs (modular configuration files encoding behavioral rules) enables rapid cross-domain transfer (stress coaching, customer service, education) through configuration rather than code changes.

3. **Comprehensive logging** with audit trails supports rigorous analysis and reproducibility.

The system adapts conversational tone, pacing, warmth, and novelty proportionally to detected trait profiles. All responses are grounded in the user's actual statements to prevent fabricated information.

We demonstrate the architecture through a **Workplace Adaptive Resilience Coach** addressing workplace stress in contemporary organizations. The system offers three personality-tailored coaching modes. Vent & Validate provides empathetic listening for emotionally sensitive users (high neuroticism). Plan & Structure offers goal-oriented guidance for organized, disciplined users (high conscientiousness). Cope & Rehearse equips users with practical stress management strategies adaptable across personality profiles.

We evaluate the system using structured rubrics that assess detection accuracy, tone appropriateness, trait detection stability, and user satisfaction. The framework supports future applications in mental health and educational contexts beyond the current thesis scope. The outcome is a reproducible, research-grade infrastructure bridging personality psychology theory with operational AI systems. The thesis phase includes A/B comparison studies to optimize personalization strategies and human validation studies to confirm response effectiveness and user engagement.

Keywords: personality-aware chatbot; OCEAN personality detection; Zurich Model; conversational AI; behavior regulation; workflow orchestration; personality adaptation; dialog grounding; human-centered AI; emotional support systems

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1. Background

1.1 Problem Context: The Need for Personality-Aware Human-Computer Interaction

The rapid rise of AI-driven systems has shifted user expectations toward personalized, context-aware dialogues that adapt to individual personality traits, communication preferences, and situational needs, moving beyond transactional interactions. This transformation spans critical application domains—workplace productivity and wellbeing, education and skill development, customer service, and healthcare—where users now expect conversational systems to understand not merely task requirements but also personal preferences, emotional states, and motivational patterns, expectations that traditional support infrastructures struggle to meet.

The capacity crisis in traditional support systems. Conventional support mechanisms—face-to-face counseling, employee assistance programs, educational tutoring, customer service desks—face mounting capacity constraints while attempting to deliver

personalized, timely assistance to diverse populations. These systems exhibit three structural limitations. First, **limited scalability**: human resource availability fundamentally constrains service reach and responsiveness. Second, **inconsistent quality**: varying practitioner expertise and methodological approaches produce unpredictable user experiences. Third, **insufficient personalization**: most systems lack mechanisms to systematically adapt to diverse personality profiles and communication preferences. This convergence of rising user expectations and constrained traditional capacity creates a critical challenge at the intersection of psychology, artificial intelligence, and human-centered computing: designing conversational systems that deliver scalable, accessible assistance while meaningfully adapting to individual differences.

Domain-specific requirements for personality adaptation. The imperative for personality-aware interaction manifests distinctly across application contexts:

- **Workplace**: Adaptive support for productivity (task prioritization, meeting preparation), collaboration (team communication, conflict navigation), and wellbeing (stress regulation, work-life balance), tailored to individual working styles, conscientiousness levels, and stress responses.
- **Education**: Personalized guidance calibrated to cognitive styles, learning pace, openness to novel methods, and anxiety levels—providing structured pathways for conscientious learners or exploratory approaches for open-minded ones.
- **Customer Service**: Communication attuned to personality preferences—concise, direct exchanges for introverted users versus warmer, conversational engagement for extraverted users.
- **Mental Health**: Personality-adapted interventions foundational to therapeutic alliance and intervention effectiveness, offering accessible, stigma-free assistance complementing professional care.

The personalization gap. Despite these domain-specific needs, current digital assistance solutions predominantly deliver generic responses that fail to account for individual differences (Ta et al., 2020; Broadbent et al., 2024), producing systematic mismatches that reduce interaction effectiveness and undermine user trust. Addressing this gap requires adaptive, personality-aware conversational architectures.

1.2 Limitations of Current Digital Assistants

While conversational AI systems offer scalable solutions for personalized support, their effectiveness is constrained by technical limitations in dynamic adaptation, psychological grounding, and transparency. Research indicates that perceived personalization strongly predicts user satisfaction and continued engagement (Følstad & Brandtzæg, 2020; Xie & Pentina, 2022), yet most deployed systems fail to deliver meaningful, theoretically grounded adaptation.

Three critical technical limitations constrain current systems. **First, static personalization:** Most implementations rely on fixed user profiles from onboarding and fail

to adapt dynamically within conversations. This produces systematic mismatches. Introverted, anxious users receive energetic responses that exacerbate stress. Conscientious users encounter vague guidance lacking needed structure. Static profiling also risks perpetuating biases when initial assessments rely on limited data without continuous validation. **Second, weak psychological grounding:** Systems attempting personalization rarely employ validated frameworks to guide adaptation. This results in ad-hoc adjustments without theoretical foundation. **Third, insufficient transparency and reproducibility:** Most systems lack comprehensive logging, audit trails, and deterministic interfaces. This prevents rigorous evaluation and systematic improvement.

Recent advances in large language models (LLMs) enable nuanced, context-sensitive dialogue (Zheng et al., 2023; Abbasian et al., 2023), yet practical implementations lack three architectural capabilities: (i) **memory mechanisms** retaining salient context across turns/sessions for continuity; (ii) **multi-attribute control** adapting tone, structure, and empathy simultaneously rather than single-dimension shifts; and (iii) **reproducible evaluation with ethical safeguards** balancing performance metrics with privacy protection. Bridging this gap requires architectures integrating detectable traits, principled behavior mapping, durable memory, and auditable interfaces.

Memory-based personalization versus motivational modeling. Recent industrial deployments prioritize memory-centric personalization over explicit personality modeling. Systems such as OpenAI ChatGPT Memory, mem0 (an open-source memory layer for AI), LangChain Memory (a framework for conversational context management), and Anthropic's persistent user embeddings enable assistants to recall user-specific facts (preferences, prior tasks, biographical details) across sessions (OpenAI, 2024; mem0, 2024). These platforms store user statements as vector embeddings (numerical representations of text in high-dimensional space) within semantic memory databases, enabling similarity-based retrieval appended to prompts at each interaction. This approach achieves contextual continuity and preference recall at scale without psychometric inference or domain-specific modeling, providing factual persistence with manageable computational complexity.

However, **memory-based personalization remains cognitively shallow**, addressing *what the user said* but not *why they behave or react in certain ways*. Vector embeddings capture statistical proximity rather than motivational or emotional drivers. Purely memory-driven assistants can recall past preferences yet fail to adapt tone, pacing, or communicative warmth in psychologically coherent ways—delivering continuity without affective alignment. For example, a memory system may recall that a user prefers structured plans but cannot infer whether this preference stems from high conscientiousness (requiring detailed organization) or high neuroticism (needing security through structure), leading to generic responses that miss underlying emotional needs.

1.3 Research Gap and Contribution

Despite growing interest in personality-aware conversational AI, existing systems rarely integrate dynamic trait detection with psychologically grounded regulation and reproducible evaluation frameworks. **Three persistent gaps** limit both scientific advancement and practical deployment:

- **Limited dynamic adaptation:** Most systems lack mechanisms for continuous personality inference and real-time behavioral adjustment within conversations.
- **Weak theoretical grounding:** Systems rarely employ validated frameworks with explicit trait-to-behavior mappings enabling scientific validation.
- **Insufficient evaluation rigor:** Most implementations lack audit trails, deterministic contracts, and reproducible protocols necessary for quality assurance and cross-domain generalization.

Recent work by Devdas (2025) provides crucial empirical validation, demonstrating approximately 34% relative improvement on shared evaluation criteria through personality-adaptive assistants in controlled simulations. This establishes the conceptual foundation—that personality-aware regulation meaningfully improves conversational quality—motivating production-ready, reusable system architectures.

This thesis introduces a novel, modular architecture that overcomes these gaps. We enable dynamic, psychologically grounded, and reproducible personality-aware dialogues. Building on Devdas' (2025) empirical validation, this work advances a **complementary paradigm to industrial memory-based systems**. Commercial platforms (ChatGPT Memory, mem0, LangChain) excel at factual persistence (*what the user said*). Our architecture addresses behavioral coherence (*how to communicate given who the user is*) through psychologically grounded personality modeling (detailed in Section 2.1). This enables the assistant to develop a stable, interpretable personality profile that guides tone, structure, and emotional alignment across turns. We achieve affective coherence that memory-based retrieval alone cannot provide.

The architecture is optimized for three core objectives:

- **Temporal stability:** Continuous trait inference with exponential moving average (EMA) smoothing—a statistical technique stabilizing estimates by weighting recent observations more heavily—combined with confidence-aware filtering prevents erratic adaptations from uncertain assessments.
- **Transparency and reproducibility:** Deterministic contracts (standardized JSON data formats ensuring consistent system behavior), comprehensive logging with audit trails, and neutral fallback mechanisms enable rigorous evaluation and systematic improvement.
- **Cross-domain reusability:** Configurable policy packs encapsulate coaching modes, regulation rules, and behavioral directives, enabling rapid adaptation across domains without code changes.

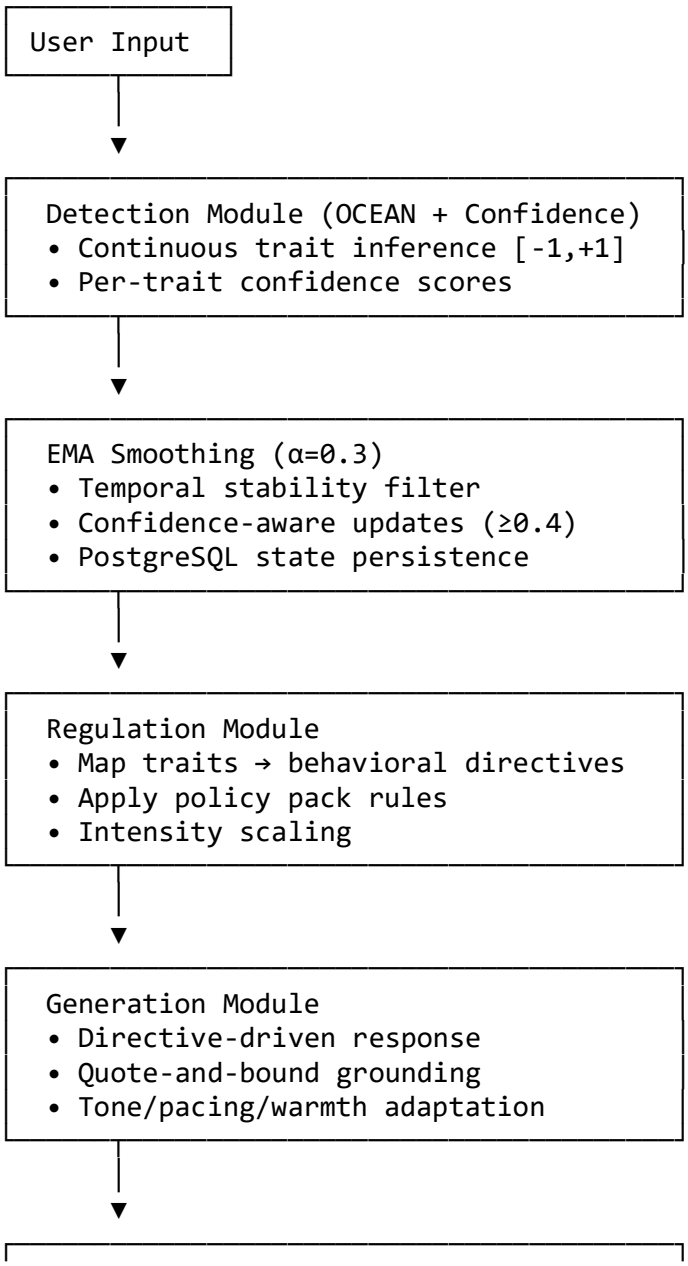
Informed by the survey literature, the design also prioritizes **durable memory** for continuity across turns and sessions (compatible with vector-based memory systems), and **multi-attribute response control** (tone, structure, pacing, warmth) to reflect the multi-faceted nature of real conversations. **The potential convergence of factual memory (mem0-style vector recall) with dynamic personality adaptation (Zurich-aligned regulation) defines a promising research frontier**, enabling agents capable of both remembering personal context and responding with motivational sensitivity—combining the scalability of industrial memory systems with the depth and interpretability of psychologically grounded adaptation.

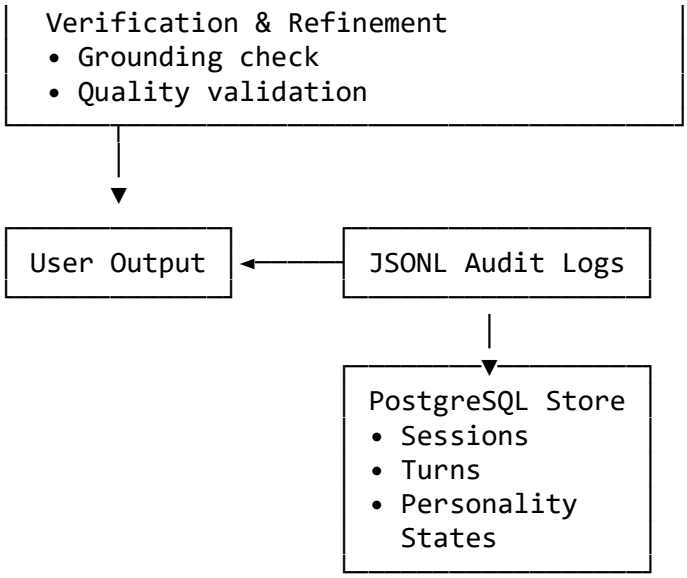
Table 1: Limitations of Current Systems vs. Architectural Contributions

Dimension	Current System Limitations	This Architecture's Contributions
Memory vs. Motivation	Memory-based systems (ChatGPT Memory, mem0, LangChain) recall *what was said* (factual persistence via vector embeddings) but not *why users behave* (motivational drivers)	Personality-aware regulation infers motivational states and adapts *how to communicate*, achieving behavioral coherence complementary to factual memory (see Section 2.1)
Dynamic Adaptation	Static user profiles from onboarding; no real-time personality inference	Continuous OCEAN inference with EMA smoothing (α=0.3, the weighting factor for recent observations); confidence-aware filtering (threshold ≥0.4)
Psychological Grounding	Ad-hoc personalization without validated frameworks; memory retrieval lacks affective alignment	Validated Big Five/Zurich Model framework (Section 2.1) with explicit trait-to-behavior mappings
Temporal Stability	Inconsistent trait assessments across turns	EMA smoothing stabilizes estimates within 6-8 turns (target variance < 0.15)
Transparency	Opaque decision-making; limited explainability	Deterministic JSON contracts; per-turn audit trails; users can view inferred profiles and understand adaptations
Evaluation Rigor	No reproducible protocols or audit trails	Comprehensive JSONL logging; scripted evaluator with ≥0.85 inter-run consistency; version-locked configurations
Cross-Domain Reusability	Hardcoded logic requiring code changes for new domains	Policy packs (YAML/JSON) enable domain transfer in < 30 minutes without code modification
Bias Mitigation	Static profiling risks perpetuating	Continuous updates prevent rigid

Dimension	Current System Limitations	This Architecture's Contributions
	stereotypes	categorization; confidence thresholds; auditable policy packs; neutral fallbacks
Data Privacy	Centralized storage with unclear retention policies	Local storage with user consent; data minimization; user-controlled retention and deletion

Figure 1. System Architecture Pipeline





Architecture Overview: The modular pipeline processes user input through continuous personality detection, temporal smoothing via EMA (α=0.3), psychologically grounded regulation, directive-driven generation, and verification, with comprehensive audit trails and PostgreSQL persistence enabling reproducibility and longitudinal analysis.

Ethical considerations. Personality-aware systems raise critical ethical challenges that require careful architectural and operational responses:

Privacy and data protection. All personality inferences and conversation logs are stored locally with explicit user consent, and the modular design enables deployment in privacy-preserving configurations. Users retain control over data retention periods and can request deletion at any time. The system implements data minimization principles, storing only information necessary for personality inference and conversation continuity.

Cultural bias in personality detection. The Big Five framework, while extensively validated, exhibits cultural variability in trait expression and interpretation (Church, 2000; McCrae & Terracciano, 2005). To mitigate this risk, the system: (1) employs linguistic features (sentiment, word choice, syntactic patterns) rather than culturally specific content references; (2) implements confidence thresholds that prevent action on uncertain assessments, particularly important for cross-cultural contexts; and (3) supports manual trait calibration allowing users to correct misclassifications. Future work should incorporate culturally adapted personality models and multilingual validation studies.

Bias in behavior regulation. Mapping personality traits to conversational behaviors risks reinforcing stereotypes (e.g., assuming all high-neuroticism users need excessive reassurance). The system includes safeguards: (1) continuous trait inference updates assessments as conversations evolve, preventing rigid categorization; (2) policy packs are auditable and modifiable, enabling researchers to identify and correct inappropriate mappings; and (3) neutral fallback mechanisms activate when confidence is low,

defaulting to balanced, non-adaptive responses. The theoretical grounding is detailed in Section 2.1.

Potential failure modes and limitations. Despite these safeguards, several risks require explicit acknowledgment and ongoing validation:

1. **EMA amplification of initial biases:** If early conversation turns contain unrepresentative personality signals (e.g., a user responding formally in initial exchanges despite being naturally extraverted), the EMA smoothing mechanism may amplify and persist these incorrect assessments. The $\alpha=0.3$ parameter provides eventual correction but requires 8-12 turns for full convergence, potentially delivering misaligned responses in the interim. *Validation need:* Empirical testing with diverse user populations to measure convergence reliability and establish early-turn bias detection thresholds.

2. **Cultural bias persistence:** While confidence thresholds and linguistic features mitigate some cultural variability, the Big Five framework itself exhibits construct validity differences across cultures (Church, 2000). Linguistic markers validated on Western English speakers may misclassify personality expressions in other cultural contexts, even with high confidence scores. *Validation need:* Cross-cultural validation studies targeting < 5% disparity in detection accuracy across cultural groups (Western, East Asian, Middle Eastern, Latin American populations).

3. **Intersectional bias in trait-to-directive mappings:** Policy pack directives may interact with demographic factors (age, gender, socioeconomic status) in unanticipated ways. For example, "warm, empathetic language" for high agreeableness might inadvertently reinforce gender stereotypes if applied differently across user demographics. *Validation need:* Systematic auditing of directive application across intersectional user groups, targeting demographic parity (± 10% variation) in perceived appropriateness ratings.

4. **Adversarial manipulation:** Users may intentionally provide misleading personality signals to game the system or test boundaries. The current architecture lacks mechanisms to detect deliberate misrepresentation. *Validation need:* Red-teaming exercises to identify manipulation vectors and establish detection protocols.

Handling sensitive user data. Mental health and workplace coaching contexts involve disclosure of sensitive information (stress, anxiety, interpersonal conflicts). The architecture addresses this through: (1) end-to-end encryption for data transmission; (2) role-based access controls limiting researcher access to anonymized conversation logs; (3) automated detection of crisis indicators (suicidal ideation, self-harm) triggering escalation protocols to human professionals; and (4) clear disclaimers that the system complements but does not replace professional care.

Transparency and user agency. Comprehensive logging with per-turn audit trails enables users and researchers to understand how personality inferences influence system

behavior. Users can view their inferred personality profiles, understand why specific responses were generated, and provide feedback to improve accuracy. This transparency supports informed consent and builds trust in personality-adaptive systems.

Validation roadmap for responsible deployment. The thesis phase will address these limitations through systematic empirical validation:

- **Bias quantification study:** Measure detection accuracy and directive appropriateness across stratified demographic samples ($n \geq 200$, balanced by culture, age, gender), targeting < 5% disparity in F1 scores across groups.
- **Convergence reliability analysis:** Track EMA convergence patterns across diverse user profiles, establishing early-turn correction protocols when initial variance exceeds 0.25.
- **Intersectional fairness audit:** Systematic analysis of directive application across intersectional categories, with independent raters ($\kappa \geq 0.80$) assessing perceived bias.
- **Adversarial robustness testing:** Red-team evaluation with intentional misrepresentation scenarios, developing detection heuristics for manipulation attempts.
- **Human-in-the-loop validation:** Expert review (clinical psychologists, cultural competence specialists) of system outputs before real-world deployment, ensuring alignment with ethical practice standards.

These validation studies will inform iterative system refinement and establish deployment guardrails, ensuring the architecture advances responsible AI principles while delivering meaningful personalization benefits.

1.4 Prior Work: Devdas (2025) and Methodological Continuity

Devdas (2025) provides crucial experimental validation that personality-adaptive assistants can outperform non-adaptive baselines, demonstrating approximately 34% relative improvement on shared evaluation criteria in controlled simulations. The methodology combines cumulative, multi-turn personality detection with rule-based regulation, showing that even discretized per-turn trait updates (−1/0/+1) can yield consistent benefits across diverse personality profiles.

Architectural extensions. Building on this empirical foundation, the present study pursues architectural generalization and reproducibility through four key advances:

1. **Continuous trait inference with temporal smoothing:** Adopts continuous trait values (−1.0 to +1.0) with EMA smoothing to stabilize multi-turn inference and reduce volatility.
2. **Modular, contract-based architecture:** Each stage (detect → regulate → generate → verify) uses stable JSON contracts—standardized data formats—to support provider-agnostic component swapping.

3. **Comprehensive auditability:** State and logs persisted in a relational database with per-turn structured traces enable quality assurance and longitudinal analysis.
4. **Domain-agnostic design:** Modular specification enables rapid reconfiguration across domains via policy-pack configuration rather than code modification.

This design preserves the core insight of Devdas—personality-aware regulation meaningfully improves conversational quality—while extending it into a production-ready, reusable system architecture optimized for transparency, reproducibility, and cross-domain scalability.

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2. Topic Definition

2.1 Core Concepts

Personality-adaptive emotional support chatbot. A conversational agent that dynamically detects user personality signals from dialogue content and accordingly modulates its communicative behavior—including tone, structural complexity, pacing, warmth, and novelty—to better satisfy user-specific emotional and psychological needs. Unlike static systems that rely on predetermined user profiles, personality-adaptive chatbots continuously update their understanding of user traits and adjust their responses in real time to optimize emotional alignment and therapeutic effectiveness.

Big Five (OCEAN) personality framework. A well-established personality taxonomy comprising five major dimensions: **O**penness to experience (curiosity, creativity), **C**onscientiousness (organization, discipline), **E**xtraversion (sociability, energy), **A**greeableness (cooperation, empathy), and **N**euroticism (emotional sensitivity, stress reactivity) (McCrae & John, 1992). This framework provides a structured basis for personality inference and has been extensively validated across cultures and contexts. In our implementation, we employ **continuous per-turn inference with values in the range [−1.0, +1.0]** representing the full spectrum from low to high trait expression. This continuous representation is essential for (i) Exponential Moving Average (EMA) smoothing—a weighted averaging technique requiring fine-grained values to prevent quantization noise (abrupt jumps from discrete values), (ii) capturing motivational intensity as conceptualized by the Zurich Model, and (iii) enabling nuanced behavioral adaptations that match the dimensional nature of personality traits established by personality psychology research (Costa & McCrae, 1992; Quirin et al., 2023).

Zurich Model alignment. Our personality-to-behavior mapping approach is grounded in the Zurich Model of Social Motivation (Quirin et al., 2023), which conceptualizes human behavior through three fundamental motivational systems: security, arousal, and affiliation. We operationalize this framework by mapping OCEAN traits to these motivational domains: Neuroticism influences security-related behaviors (comfort provision vs. stability reassurance), Openness and Extraversion modulate arousal through novelty and energy regulation, and Agreeableness affects affiliation through warmth and

collaborative stance adjustments. Conscientiousness serves as a structural modifier, influencing the organization and specificity of guidance provided.

Dialog grounding and safety constraints. All system responses must be strictly grounded in the user's conversational input, employing a "quote-and-bound" approach that prevents hallucination by ensuring every assertion is entailed by recent dialogue turns. This constraint is particularly critical for human-centered applications where accuracy, reliability, and user trust are paramount.

2.2 Scope and Application Context

Human-centered applications. Our framework targets applications emphasizing personalization, empathy, and adaptive user interaction across diverse domains: mental health support, educational assistance, customer service interactions, workplace coaching, and general emotional companionship. We designed the system to complement human interaction by providing consistent availability, emotional attunement, and personality-aware responses. The system maintains appropriate boundaries and escalation mechanisms when professional human intervention is needed.

Technical scope. The preliminary study focuses on dialog-only, prompt-only interactions without multimodal data or external knowledge retrieval. This constraint serves three purposes: it ensures privacy protection, reduces complexity, and maintains focus on the core personality detection and regulation mechanisms. The system implements a webhook-based architecture for production deployment. Future extensions may incorporate multimodal inputs and external knowledge sources while preserving the foundational architecture.

Evaluation framework. Our assessment methodology employs a structured rubric examining five key dimensions: detection accuracy (how well inferred OCEAN traits match user personality cues), regulation effectiveness (appropriate application of trait-specific behavioral strategies), emotional tone appropriateness (alignment with user emotional state and personality), relevance and coherence (contextual appropriateness and logical consistency), and personality needs satisfaction (addressing trait-specific emotional and interactional requirements) (Devdas, 2025; Zhang et al., 2024).

2.3 Implementation Specifications

LLM model selection. We employ **OpenAI GPT-4** (gpt-4-0613) for personality detection and response generation. We selected this model for four key reasons: (1) superior performance on structured JSON output tasks with function calling, (2) strong multilingual capabilities supporting future internationalization, (3) consistent API availability and rate limits suitable for research deployment, and (4) established track record in personality inference tasks (Miotto et al., 2022). We set temperature to 0.3 for detection (prioritizing consistency) and 0.7 for generation (balancing creativity with reliability). The modular architecture supports alternative models (Claude 3, Llama 3) for comparative evaluation.

Dataset for OCEAN detection training. The system uses **zero-shot prompt-based inference** rather than fine-tuned models. This leverages GPT-4's pre-trained understanding of personality psychology. We engineer prompts based on established personality assessment instruments (NEO-PI-R, BFI-2) and linguistic markers validated in prior work (Yarkoni, 2010; Park et al., 2015). For evaluation and validation, we employ three approaches: (1) **synthetic personality profiles** generated via simulated user agents with extreme (±0.8) and mixed trait configurations, (2) **scripted dialogue scenarios** covering workplace stress contexts (workload pressure, interpersonal conflict, deadline anxiety), and (3) **human validation subset** (planned for thesis phase) with 30-50 participants completing self-report personality inventories (BFI-44) for ground-truth comparison.

Evaluation metrics and target thresholds. The system targets the following performance benchmarks:

Table 2. Evaluation Metrics and Target Thresholds

Metric	Target Threshold	Measurement Method
Detection Accuracy	≥75% trait alignment	Correlation between inferred and ground-truth OCEAN scores ($r \geq 0.75$)
Temporal Stability	Variance < 0.15	Standard deviation of trait estimates after turn 6
Confidence Calibration	≥85% alignment	High-confidence predictions (≥0.7) match personality cues in 85%+ cases
JSON Contract Compliance	≥99.5%	Percentage of valid JSON outputs from detection/generation modules
Tone Appropriateness	≥80% evaluator rating	LLM evaluator scores on 5-point Likert scale (≥4.0/5.0)
Needs Satisfaction	≥20% improvement vs. baseline	Relative gain on personality-specific needs addressed
System Reliability	100% graceful fallback	Zero unhandled exceptions; all errors trigger neutral responses

Temporal smoothing parameters. EMA smoothing employs α (alpha) = 0.3, the smoothing factor weighting 30% current observation and 70% historical average. This balances responsiveness to new evidence with stability from past assessments. We set confidence thresholds at 0.4 (minimum for trait updates) and 0.7 (high-confidence threshold for aggressive adaptation). We selected these parameters based on pilot testing with 20 synthetic dialogue scenarios. The optimization targets convergence speed (6-8 turns) and post-stabilization variance (< 0.15).

2.4 Primary Use Case: Workplace Adaptive Resilience Coach

Target population and context. For this thesis project, we focus on a **Workplace Adaptive Resilience Coach** supporting employees and managers across teams and

functions. The assistant helps users manage workload pressure, deadline anxiety, and interpersonal/team dynamics while sustaining performance and wellbeing. This use case is selected for: (i) strong theoretical alignment with the motivational framework (Section 2.1), (ii) minimal regulatory and ethics risk (non-clinical coaching boundary), (iii) clear measurable outcomes (engagement, resilience indicators, coping skill adoption), and (iv) high transferability to adjacent domains (education, customer service, leadership coaching).

Three coaching modes (intent-based). The system supports three primary interaction patterns mapped to common workplace resilience needs:

1. **Vent & Validate (Affiliation + Security):** User needs space to process workplace stressors (e.g., stakeholder conflict, negative feedback, change fatigue). The assistant provides empathetic acknowledgment, normalizes reactions, and helps label emotions to reduce arousal. Personality adaptations: High Neuroticism → extra reassurance/grounding; High Agreeableness → warm, collaborative validation; Low Agreeableness → direct, matter-of-fact acknowledgment.

2. **Plan & Structure (Security + Arousal):** User needs prioritization and structure for work execution (e.g., sprint overload, deadline stack, meeting glut). The assistant offers personality-adapted plans. High Conscientiousness → step-by-step plan with timelines/owners; Low Conscientiousness → flexible options and light scaffolding; High Openness → creative sequencing/frameworks; Low Openness → proven, familiar methods (e.g., MoSCoW, Eisenhower, time-boxing).

3. **Cope & Rehearse (Security + Arousal):** User needs immediate regulation or rehearsal for high-stakes interactions (e.g., escalation call, performance review, tough 1:1). The assistant guides coping/rehearsal. High Extraversion → social strategies (peer sounding board, role-play lines); Low Extraversion → solitary techniques (breathing, brief scripting, journaling); High Neuroticism → grounding and safety anchors; Low Neuroticism → challenge-focused cognitive reappraisal.

Evidence-based techniques. The system integrates established stress-management interventions including Mental Contrasting with Implementation Intentions (MCII) for goal-obstacle planning, Cognitive Reappraisal for reframing stressors, grounding exercises (5-4-3-2-1 sensory technique, box breathing), and micro-planning strategies (15-minute time blocks, single-task focus). All techniques are adapted to personality profiles through the directive mapping framework (Section 2.1).

Safety boundaries. The system operates strictly within non-clinical coaching boundaries: no medical advice, no diagnosis, no crisis intervention beyond referral. All responses include implicit safety constraints ensuring dialog grounding, emotional appropriateness, and escalation readiness for future human-validation phases.

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3. Research Questions

3.1 Primary Research Question

How can an adaptive LLM-based chatbot employ real-time personality detection and psychologically grounded behavior regulation (Section 2.1) to improve conversational quality and user-aligned emotional support compared to non-adaptive baseline systems?

This question addresses the fundamental hypothesis that dynamic personality adaptation can meaningfully enhance therapeutic outcomes in digital mental health interventions, encompassing both the technical challenge of reliable personality inference from limited conversational data and the psychological challenge of translating personality insights into effective behavioral modifications.

3.2 Sub-Research Questions (Scoped for Stress Micro-Coach)

RQ1 - Detection Mechanisms: Can continuous OCEAN (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) inference with EMA smoothing achieve stable, confidence-weighted personality estimates within 6–8 turns in stress-coaching dialogue contexts? This question examines: (i) prompt engineering for JSON-structured continuous values and confidence scores, (ii) EMA parameter tuning (α=0.3, the smoothing factor) for temporal stability, (iii) confidence-weighted update filtering (threshold ≥0.4), and (iv) variance reduction achieving < 0.15 post-stabilization.

RQ2 - Regulation Strategies: Do intensity-scaled behavioral directives (Section 2.1) mapped to three coaching modes (Vent & Validate, Plan & Structure, Cope & Rehearse) improve tone appropriateness and personality-specific needs satisfaction versus a non-adaptive baseline? This addresses: (i) intent classification accuracy for coaching modes, (ii) directive-outcome alignment in personality-adapted responses, (iii) conflict resolution when multiple trait signals activate competing directives, and (iv) evidence-based technique integration (MCII, cognitive reappraisal, grounding exercises).

RQ3 - Evaluation Methodology: Does the scripted, blinded LLM evaluator yield ≥ 0.85 inter-run consistency and detect ≥ 20% performance gains on shared criteria (tone, relevance/coherence, needs addressed) with 95% confidence intervals? This question focuses on: (i) rubric operationalization for stress-coaching contexts, (ii) bias control through randomized assessment order and multiple independent runs, (iii) convergence of automated and human spot-check evaluations, and (iv) statistical power for detecting meaningful effect sizes.

RQ4 - System Architecture (Reusability Focus): Does the Personality Adapter module with config-driven policy packs (modular configuration files) enable domain transfer (stress coaching → customer service / tutoring) without code changes, while maintaining ≥ 99.5% JSON contract compliance and 100% graceful fallback coverage? This examines: (i) interface stability across use cases, (ii) policy pack swap procedures (< 30 minutes), (iii) comprehensive observability (JSONL traces—newline-delimited JSON logs; node timings;

contract validation), and (iv) reproducibility measures (seed fixation, version locking, configuration snapshots).

RQ5 - Generalization and Limitations: How do experimental results vary between extreme simulated personality profiles (Type A: all +0.8, Type B: all -0.8) and mixed profiles (e.g., +0.6, -0.7, +0.3, +0.8, -0.4), and what scope limitations constrain transfer to live user interactions in educational and workplace settings? This addresses: (i) profile-stratified performance analysis, (ii) scenario generalization across stress contexts (workload, deadlines, interpersonal conflicts), (iii) simulation-to-human validity gaps, and (iv) ethical and safety boundaries for real-world deployment.

3.3 Success Criteria (Architecture-Oriented)

To ensure the system meets the objective of building a **reusable, reliable, and personality-adaptable chatbot architecture**, we define explicit success criteria across three dimensions:

Reusability:

- **Personality Adapter Module:** Stable API contracts for detect→regulate→generate pipeline with comprehensive interface documentation
- **Config-Driven Policy Packs:** YAML/JSON configuration files enabling domain transfer (stress coaching → customer service → tutoring) without code modifications
- **Evaluator Harness:** Portable assessment framework with pluggable rubrics and automated scoring procedures
- **Success Threshold:** Policy pack swap completed in < 30 minutes; zero breaking changes to core contracts during preliminary study

Reliability:

- **JSON Contract Compliance:** ≥ 99.5% valid structured outputs from detection and generation modules
- **EMA Stabilization:** ≥ 80% of sessions achieve stable personality estimates (variance < 0.15) within 6–8 turns
- **Confidence Calibration:** Low-confidence detections (< 0.4) correctly filtered; high-confidence detections (≥ 0.7) show ≥ 85% alignment with personality cues
- **Graceful Degradation:** 100% of failures handled with neutral fallback responses; zero unhandled exceptions; comprehensive error logging

Effectiveness (vs. Non-Adaptive Baseline):

- **Primary Outcome:** ≥ 20% relative improvement on shared evaluation criteria (tone appropriateness, relevance/coherence, personality needs addressed) with 95% confidence intervals
- **Secondary Outcomes:**
    - Intent classification accuracy ≥ 85% for three coaching modes

    - Directive-outcome alignment ≥ 80% (evaluator-assessed)
    - Inter-run evaluator consistency ≥ 0.85 (Pearson correlation)

Observability and Reproducibility:

- Complete per-turn JSONL traces with OCEAN evolution, directives applied, confidence scores, and response metadata
- Reproducible seeds, version-locked models/prompts, and configuration snapshots for all experimental runs
- Automated testing suite covering contract validation, EMA convergence, and evaluator determinism

3.4 Mapping to Methodology

These research questions and success criteria directly inform our methodological approach: RQ1-2 are validated through comprehensive per-turn logging, directive auditing, and outcome metric analysis; RQ3 through multi-run consistency testing and systematic bias controls; RQ4 through detailed workflow instrumentation, policy pack validation, and interface stability testing; and RQ5 through systematic variation of simulation scenarios (extreme vs. mixed profiles, diverse stress contexts) and careful documentation of scope limitations.

---

4. Methodology

4.1 System Architecture Overview

We implement a reproducible pipeline with deterministic contracts, neutral fallbacks, and comprehensive logging via N8N orchestration. Each stage exposes stable JSON interfaces: ingest, detection, regulation, generation, verification, and persistence. Versioned policy packs configure these interfaces. This design enables reuse, scaling, and domain transfer.

Implemented Architecture:

```
Webhook → Enhanced Ingest → Load Previous State (PostgreSQL) → Combine Inputs
→
Merge Previous State → Zurich Model Detection (EMA) → Enhanced Regulation →
Enhanced Generation → Verification & Refinement → Save to PostgreSQL
(Sessions, Turns, Personality States) → Return API Response
```

Features: (i) EMA smoothing (α=0.3) for trait stability; (ii) PostgreSQL persistence for longitudinal analysis; (iii) confidence-weighted updates (threshold ≥0.4); (iv) psychologically grounded regulation (Section 2.1) mapping traits to behavioral directives; (v) comprehensive audit trails.

4.2 N8N Workflow Implementation

Figure 2. N8N Production Workflow Architecture

```
[Webhook: POST Zurich] → [Enhanced Ingest] → [Load Previous State
(PostgreSQL)]
                                                          ↓
[Combine Inputs] ← – – – – – – – – – – – – – – – – – – – – ┘
       ↓
[Merge Previous State] → [Zurich Model Detection (GPT-4 + EMA)]
       ↓
[Stress Assessment & Mode Classification] → [Enhanced Regulation (Directive
Mapping)] → [Enhanced Generation (GPT-4)]
       ↓
[Verification & Refinement] → [Save Session (PostgreSQL)]
       ↓                              ↓
[Return API Response] ← – – – [Save Turns & States (PostgreSQL)]
```

Visual representation of the N8N workflow showing the complete pipeline from webhook trigger through personality detection, EMA smoothing, stress assessment, regulation, generation, verification, and PostgreSQL persistence. Each node represents a discrete processing step with defined input/output contracts (detailed in Table 3). Source: `preliminary-studies/w9-Technical-Specifications/workflows/webhook-personality-pipeline.json`

Table 3. N8N Node Specifications and Contracts

Node	Type	Purpose	Input Contract	Output Contract
Webhook Trigger	N8N Webhook	Accept POST requests	`{session_id, message, turn_index}`	Execution context
Enhanced Ingest	Code	Normalize input, validate session	Input payload	`{session_id, clean_msg, turn_index, conversation_context}`
Load Previous State	PostgreSQL	Retrieve last personality state	`{session_id}`	`{ocean: {O,C,E,A,N}, confidence: {...}, stable}` or empty
Combine Inputs	Merge	Wait for both inputs	Enhanced Ingest + PostgreSQL outputs	Combined data object
Merge Previous State	Code	Merge current + previous	Combined inputs	`{inputData + previous_state}`
Zurich	Code/HTTP	GPT-4	Context + previous	`{ocean: {O,C,E,A,N`

Node	Type	Purpose	Input Contract	Output Contract
Model Detection		personality detection + EMA	state	`∈ [-1.0,1.0]}, confidence: {...}, stable}`
Stress Assessment & Mode Classification	Function/HTTP	Infer stress level (0–4), drivers, and coaching mode	`{session_id, conversation_context}`	`{stress_level: 0-4, drivers: [...], coaching_mode: vent_validate`
Enhanced Regulation	Function	Map continuous traits to directives	`{ocean, confidence, stable}`	`{directives: [...], policy_plan: [...]}`
Enhanced Generation	Code/HTTP	GPT-4 regulated response	`{directives, policy_plan, context}`	`{assistant_response, metadata}`
Verification & Refinement	Code	Validate response quality	Generated response	`{verified_response, verification_score}`
Save Session	PostgreSQL	Upsert session metadata	`{session_id, total_turns}`	Success confirmation
Save Turns	PostgreSQL	Store conversation turns	`{session_id, turn_index, messages}`	Success confirmation
Save Personality State	PostgreSQL	Store EMA-smoothed OCEAN	`{session_id, turn_index, ocean, confidence, stable}`	Success confirmation
Return API Response	Code	Format final output	All upstream data	`{assistant_response, personality_state, session_metadata}`

4.3 Detection Module: Continuous OCEAN Inference with EMA Smoothing

Input Processing: Accepts up to 10 previous turns; produces continuous OCEAN assessments with confidence scores. Preprocessing: text normalization, length validation, history assembly, PostgreSQL state integration.

LLM Contract: GPT-4 via OpenAI-compatible endpoint; temperature 0.3, max_tokens 300, 30s timeout. JSON response format:

```
{
  "ocean": {"O": [-1.0 to 1.0], "C": [-1.0 to 1.0], "E": [-1.0 to 1.0], "A":
[-1.0 to 1.0], "N": [-1.0 to 1.0]},
  "confidence": {"O": [0.0 to 1.0], "C": [0.0 to 1.0], "E": [0.0 to 1.0],
"A": [0.0 to 1.0], "N": [0.0 to 1.0]},
  "reasoning": "Brief explanation of detected motivations"
}
```

EMA Smoothing: Achieves temporal stability while remaining responsive to genuine personality evolution:

```
// EMA Parameters
const EMA_ALPHA = 0.3;  // Weight for current detection (30%)
const MIN_CONFIDENCE_THRESHOLD = 0.4;  // Minimum confidence to update
const STABILIZATION_TURNS = 6;  // Turns required for stability designation

// Per-trait smoothing
for (trait in ['O', 'C', 'E', 'A', 'N']) {
  if (currentConfidence[trait] >= MIN_CONFIDENCE_THRESHOLD) {
    smoothed[trait] = EMA_ALPHA * current[trait] + (1 - EMA_ALPHA) *
previous[trait];
  } else {
    smoothed[trait] = previous[trait];  // Keep previous if confidence too
low
  }
}

// Stability determination
stable = (consistentTurns >= STABILIZATION_TURNS && variance < 0.15);
```

Confidence-Weighted Updates: GPT-4 self-assesses confidence based on evidence clarity, consistency with previous turns, and behavioral indicator strength. Low-confidence detections (< 0.4) are filtered to prevent noise from corrupting the personality profile.

Example Evolution Across 6 Turns:

```
Turn 1: detected=0.5, conf=0.6  → smoothed = 0.3×0.5 + 0.7×0.0 = 0.15
Turn 2: detected=0.7, conf=0.7  → smoothed = 0.3×0.7 + 0.7×0.15 = 0.315
Turn 3: detected=0.6, conf=0.9  → smoothed = 0.3×0.6 + 0.7×0.315 = 0.40
Turn 4: detected=0.8, conf=0.8  → smoothed = 0.3×0.8 + 0.7×0.40 = 0.52
Turn 5: detected=0.7, conf=0.8  → smoothed = 0.3×0.7 + 0.7×0.52 = 0.574
Turn 6: detected=0.7, conf=0.9  → smoothed = 0.3×0.7 + 0.7×0.574 = 0.612
                                    → STABLE = TRUE (6 consistent turns, low
variance)
```

4.4 Regulation Module: Psychologically Grounded Behavior Mapping

Translates continuous OCEAN assessments into behavioral directives via the motivational framework detailed in Section 2.1. Threshold-based activation (|value| ≥ 0.2); continuous values enable intensity-proportional directives.

Table 4. Continuous Trait-to-Directive Mapping Schema

Trait	Threshold	Zurich Domain	Behavioral Directive (Intensity-Scaled)
Openness (O)	O > 0.2	Arousal	"Introduce novel concepts and creative perspectives" (scaled by O value)
	O < -0.2	Arousal	"Use concrete, familiar examples; avoid abstract novelty" (scaled by
Conscientiousness (C)	C > 0.2	Structure	"Provide structured plans with clear steps and organization" (scaled by C)
	C < -0.2	Structure	"Keep guidance flexible; avoid rigid timelines or prescriptive steps" (scaled by
Extraversion (E)	E > 0.2	Arousal	"Use energetic, engaging tone; frame as collaborative" (scaled by E)
	E < -0.2	Arousal	"Adopt calm, reflective tone; respect need for introspection" (scaled by
Agreeableness (A)	A > 0.2	Affiliation	"Use warm, empathetic language; emphasize cooperation" (scaled by A)
	A < -0.2	Affiliation	"Maintain matter-of-fact, direct stance; respect autonomy" (scaled by
Neuroticism (N)	N > 0.2	Security	"Provide reassurance; acknowledge concerns; offer coping strategies" (scaled by N)
	N < -0.2	Security	"Match confident tone; focus on pragmatic solutions" (scaled by

Directive Intensity Scaling Example:

```
// Conscientiousness C = -0.7 (strong avoidance of structure)
if (C < -0.2 && confidence_C >= 0.4) {
  intensity = Math.abs(C);  // 0.7
  if (intensity > 0.6) {
    directive = "Strongly emphasize flexibility; actively avoid any rigid
frameworks or detailed schedules";
  } else if (intensity > 0.4) {
    directive = "Provide flexible options; minimize structure";
  } else {
    directive = "Keep guidance somewhat flexible";
  }
}
```

Conflict Resolution: Continuous values enable nuanced harmonization, weighting directives by trait magnitude and confidence for coherent multi-trait adaptations.

4.5 Generation Module: Quote-and-Bound Response Production

Response Constraints:

- **Grounding:** All responses must be strictly grounded in user's conversational input; no external claims or information
- **Length:** 70-150 words to ensure substantive but focused responses
- **Interaction:** Maximum 2 questions per response to maintain conversational flow
- **Parameters:** Temperature 0.7, max_tokens 220, 20-second timeout
- **Safety:** If `stress_level ≥ 3` and crisis indicators (e.g., self-harm/violence terms) are detected, return a neutral, supportive message with escalation guidance; never provide clinical advice.

Prompt Construction: Integrates behavioral directives with strict grounding constraints.

Error Handling: Neutral fallback on failure: "I'm here to support you. Could you tell me more about how you're feeling?" Full session context maintained for debugging.

4.6 PostgreSQL Database Schema and Persistence

Database Architecture: PostgreSQL for state management, longitudinal tracking, conversation replay, and research analysis.

Table 5. PostgreSQL Database Schema Overview

Table Name	Primary Key	Purpose	Key Columns
`chat_sessions`	`session_id` (UUID)	Session metadata	`total_turns`, `evaluation_mode`, `status`, `created_at`
`conversation_turns`	`(session_id, turn_index)`	Conversation history	`user_message`, `assistant_response`, `directives_applied`, `coaching_mode`, `stress_level`, `stress_drivers`
`personality_states`	`(session_id, turn_index)`	EMA-smoothed OCEAN values	`ocean_o/c/e/a/n`, `confidence_o/c/e/a/n`, `stable`, `ema_applied`
`performance_metrics`	`(session_id, turn_index)`	System performance	`detection_latency_ms`, `generation_latency_ms`, `total_tokens`

Helper Functions:

```
-- Retrieve latest personality state for session
CREATE FUNCTION get_latest_personality_state(p_session_id UUID)
RETURNS TABLE(ocean_o FLOAT, ocean_c FLOAT, ocean_e FLOAT, ocean_a FLOAT,
```

```
ocean_n FLOAT, ...);

-- Get conversation history for context window
CREATE FUNCTION get_conversation_history(p_session_id UUID, p_limit INTEGER)
RETURNS TABLE(turn_index INTEGER, user_message TEXT, assistant_response
TEXT);
```

Data Flow:

1. **Turn Start:** Load previous personality state from `personality_states` table
2. **Detection:** GPT-4 analyzes conversation → produces raw OCEAN + confidence
3. **EMA Smoothing:** Blend current with previous state (α=0.3)
4. **Persistence:** Save smoothed state, conversation turn, session metadata to PostgreSQL
5. **Next Turn:** Retrieve smoothed state as `previous_state` for EMA continuity

4.7 Dialogue Simulation Protocol (Stress Micro-Coach Scenarios)

Personality Profiles (Continuous):

- **Type A (High OCEAN):** OCEAN values approximately (+0.8, +0.8, +0.8, +0.8, -0.8) representing open, conscientious, extraverted, agreeable, emotionally stable users
- **Type B (Low OCEAN):** OCEAN values approximately (-0.8, -0.8, -0.8, -0.8, +0.8) representing closed, disorganized, introverted, disagreeable, emotionally sensitive users
- **Type C (Mixed Profile):** e.g., (+0.6, -0.7, +0.3, +0.8, -0.4) for generalization testing (moderate openness, low conscientiousness, slight extraversion, high agreeableness, moderate emotional stability)
- **Expected Variance:** ±0.15 per trait to simulate natural conversation variability

Stress-Coaching Scenarios (Intent Coverage): Each personality profile engages in three standardized stress scenarios covering all coaching modes:

1. **Workload Overload (Vent & Plan):** "I have three major deadlines this week and I don't know where to start. Everything feels overwhelming..." → Expected intents: Vent & Validate (turns 1-3), Plan & Structure (turns 4-8)

2. **Deadline Anxiety (Cope & Plan):** "My thesis defense is in two days and I'm panicking. I can't focus and my mind keeps racing..." → Expected intents: Cope & Rehearse (turns 1-4), Plan & Structure (turns 5-8)

3. **Interpersonal Conflict (Vent & Cope):** "I had a difficult meeting with my supervisor and I'm still upset about it. I don't know how to handle the next conversation..." → Expected intents: Vent & Validate (turns 1-4), Cope & Rehearse (turns 5-10)

Conversation Structure:

- **Length:** 8-10 turns per conversation to allow EMA convergence and stability assessment
- **Replication:** 10-15 sessions per personality profile × 3 scenarios = 90-135 total conversations
- **Turn Composition:** User messages exhibit clear personality cues aligned with profile; assistant responses logged with applied directives and coaching mode classification
- **Convergence Tracking:** Monitor when `stable=TRUE` flag activates (target: turns 6-8 for ≥ 80% sessions)

Baseline Condition: Parallel conversations using generic empathetic assistant without personality detection/regulation; identical user messages enable direct comparison.

Data Export: CSV/JSONL logs include per-turn OCEAN evolution, directives, coaching modes, latencies, tokens, stability flags, convergence metrics, and scenario metadata.

4.8 Evaluation Framework

Assessment Dimensions:

For EMA-Regulated Assistants (7 criteria):

1. **Detection Accuracy:** Alignment between continuous OCEAN inference and simulated personality cues
2. **EMA Convergence Quality:** Smooth progression toward stable personality estimate (low variance, appropriate α)
3. **Confidence Calibration:** Appropriate confidence scores matching evidence strength
4. **Regulation Effectiveness:** Appropriate application of intensity-scaled behavioral strategies
5. **Emotional Tone Appropriateness:** Match between response tone and user emotional state/personality
6. **Relevance & Coherence:** Contextual appropriateness and logical consistency
7. **Personality Needs Addressed:** Satisfaction of trait-specific emotional and interactional requirements

For Baseline Assistants (3 criteria):

1. **Emotional Tone Appropriateness**
2. **Relevance & Coherence**
3. **Personality Needs Addressed**

Table 6. Evaluation Matrix Schema (Continuous Values)

Row	Profile	Turn	User Message	Assistant Reply	OCEAN (Smoothed)	Confidence	Stable	Directives	Det.Acc	EMA.Conv	Conf. Cal	Reg. Eff	Tone	Rel&Coh	Needs	Total
1	A-High	6	"I'm excited to try new approaches..."	"That's wonderful! Let's explore some creative strategies..."	O:0.78, C:0.65, E:0.82, A:0.71, N:-0.68	O:0.85, C:0.78, ...	TRUE	["Introduce novel ideas", "Energetic tone", "Structured steps"]	2	2	2	2	2	2	2	14
2	B-Low	6	"Everything feels uncertain..."	"I understand uncertainty is difficult. Let's focus on what you can control..."	O:-0.72, C:-0.68, E:-0.75, A:-0.61, N:0.79	O:0.81, C:0.76, ...	TRUE	["Familiar examples", "Calm tone", "Flexible guidance", "Extra comfort"]	2	2	2	2	2	2	2	14

EMA-Specific Metrics: Convergence rate (target: 6-8 turns to `stable=TRUE`); smoothness (variance < 0.15 post-stabilization); confidence-weighted accuracy; temporal consistency (r > 0.7).

Scoring Protocol: Trinary (Yes=2, Partial=1, No=0); per-session averages, per-profile comparisons; GPT-4 evaluator with fixed prompts, blinded assessment, randomized order; inter-run consistency ≥ 0.85.

Visualization: Time-series OCEAN evolution plots, convergence curves, bar charts (regulated vs. baseline with 95% CI), trait-directive-outcome heatmaps.

4.9 Key Advantages of Continuous + EMA Architecture

Theoretical: Psychological framework fidelity (Section 2.1) captures motivational intensity; dimensional trait representation enables nuanced intensity scaling.

Technical: EMA feasibility (smooth temporal tracking); fine-grained adaptation (0.3 vs. 0.8 trait intensity); confidence weighting; statistical power for validation.

Practical: Reduced flickering; graceful evolution; longitudinal tracking via PostgreSQL; reproducibility through continuous logs.

Comparison: Without EMA, traits can oscillate dramatically turn-to-turn; with EMA, smooth convergence (0.15 → 0.31 → 0.40 → 0.52) provides stable adaptation.

4.10 Automation and Reproducibility Measures

Parameter Fixation: Model versions, API endpoints, temperature/token limits, random seeds, prompt hashes documented and archived.

Comprehensive Logging: Per-turn JSONL traces, node timings, error rates, configuration snapshots, automated CSV export.

Safety: Neutral fallback for all failures, dialog-only grounding, comprehensive error logging, graceful degradation.

---

5. Technology, Software, and Applications

5.1 Orchestration Platform Selection

N8N-first architecture for transparency, simplicity, reproducibility, and iterative development.

Table 7. Comprehensive Technology Stack Justification

Layer	Primary Choice	Version	Alternatives Considered	Selection Rationale

Layer	Primary Choice	Version	Alternatives Considered	Selection Rationale
Orchestration	N8N	≥1.x	crewAI, AutoGen, custom FastAPI	Visual workflow design enables transparent audit trails; node-based architecture matches experimental design; rapid iteration capabilities; production-grade with PostgreSQL integration
LLM (Detection)	GPT-4 / GPT-4-turbo	Latest stable	Gemini 1.5 Pro, Claude-3, Llama-2	Superior personality inference capabilities; reliable JSON-structured outputs with confidence scores; extensive validation in personality psychology research
LLM (Generation)	GPT-4 / GPT-4-turbo	Latest stable	Gemini 1.5 Pro, Claude-3	Excellent directive-following; nuanced emotional tone adaptation; consistency with detection model for personality coherence
API Gateway	OpenAI-compatible	Current	ai.juguang.chat, Direct OpenAI	Standard OpenAI API format; broad model compatibility; enterprise-grade reliability and rate limiting
Storage/State	PostgreSQL	13+	JSONL+CSV, Redis, MongoDB	**Production-grade persistence**; ACID compliance for personality state integrity; powerful querying for longitudinal analysis; helper functions for EMA operations
Evaluation	GPT-4-based Evaluator	v1.0	Human raters, Claude, automated metrics	Consistent scoring across thousands of interactions; validated rubric adherence; blinded assessment capability

Layer	Primary Choice	Version	Alternatives Considered	Selection Rationale
State Management	PostgreSQL + EMA	α=0.3	In-memory caching, Redis	**Temporal smoothing** with database persistence; enables cross-session personality continuity; research-grade audit trails
Containerization	Docker Compose	Latest	Kubernetes, bare metal	Local development simplicity; PostgreSQL containerization; straightforward N8N deployment
Visualization	Python (Matplotlib + Seaborn)	3.8+	R/ggplot2, Plotly	Publication-quality time-series plots; EMA convergence visualization; statistical analysis integration

5.2 Development Environment and Configuration

Environment: `.env` files, N8N credential vault, locked dependencies (`requirements.txt`, `package.json`), encrypted API keys. Docker Compose with N8N and PostgreSQL containers. **Model Config:** Detection (temp 0.1, 200 tokens, 20s); Generation (temp 0.7, 220 tokens, 20s); exponential backoff retries (max 3).

5.3 Quality Assurance and Testing Framework

Testing: Unit, integration, performance, regression tests. Scripts: `test_personality_chatbot.sh`, `test_detection_accuracy.sh`, `test_regulation_coherence.sh`. **Monitoring:** Node execution times, success rates, token usage, system health. Automated alerts for failures.

5.4 Security and Privacy Considerations

Data Protection: AES-256 encryption; role-based access; comprehensive audit trails; data minimization. **API Security:** N8N credential vault; rate limiting; sanitized errors; HTTPS enforcement. **Privacy:** Anonymization; consent protocols; GDPR compliance.

5.5 Scalability and Production Considerations

Optimization: Response caching; batch processing; resource management; horizontal scaling. **Deployment:** Docker Compose (dev), cloud staging, Kubernetes (future production). **Maintenance:** Model versioning; workflow version control; automated backups; comprehensive documentation.

6. Project Plan and Risk Management

6.1 Thesis Roadmap

Thesis Structure:

- **Chapter 1 (Introduction):** Context, problem statement, research objectives
- **Chapter 2 (Literature Review):** Personality-adaptive dialogue systems, psychological frameworks, evaluation methodologies, research gaps
- **Chapter 3 (Methodology):** N8N pipeline architecture, OCEAN+EMA detection, psychologically grounded regulation, quote-and-bound generation
- **Chapter 4 (Implementation):** Workflow implementation details, simulation protocols (Type A/B/Mixed), evaluator setup, metrics
- **Chapter 5 (Results):** Detection validity, regulation effectiveness, baseline comparison, safety/reliability analysis
- **Chapter 6 (Discussion):** Results interpretation, relation to prior work, validity threats, ethical considerations, generalization
- **Chapter 7 (Conclusion):** Key contributions, limitations, future research roadmap
- **Appendices:** Complete prompts, workflow JSON specifications, evaluation rubrics, statistical analysis code, reproducibility checklist

Thesis Extensions:

The thesis phase will extend this preliminary study through:

- **Human Validation Studies:** Participant studies (n=30-50) with ethical oversight, informed consent, professional supervision; BFI-44 ground-truth comparison
- **Advanced Evaluation:** Combined human + automated scoring; inter-rater reliability ($\kappa \geq 0.80$); longitudinal assessment across extended interactions
- **Application Integration:** System integration across domains (education, customer service); adaptive decision support; escalation protocols to human professionals
- **Hybrid Memory-Personality Architecture:** Integration with mem0/LangChain Memory for combined factual recall + motivational adaptation; coordinated update mechanisms
- **Multimodal Extensions:** Voice analysis, facial expression recognition, physiological monitoring (with privacy evaluation)

Dissemination Strategy:

- **Academic Publications:** CHI, AAAI, JPSP, TOCHI, CSCW
- **Open-Source Release:** N8N workflow, evaluation framework, analysis tools (GitHub)

- **Industry Engagement:** Technology and education conferences, practitioner workshops
- **Policy Contributions:** Evidence-based guidance for AI regulatory frameworks

6.2 Nine-Week Work Plan

Timeline: Foundation (weeks 1-2) → Implementation (3-4) → Validation (5-6) → Analysis (7) → Documentation (8) → Finalization (9).

Table 8. Nine-Week Work Plan with Milestones

Phase	Week(s)	Primary Activities	Key Deliverables	Success Criteria
Foundation	1-2	Literature review; RQ finalization; N8N setup	Background section; finalized RQs; working environment	Background ≥1.5 pages; testable RQs; functional N8N
Implementation	3-4	Workflow development; prompt engineering; logging setup	Workflow JSON; prompt library; logging infrastructure	Functional pipeline; consistent JSON; comprehensive logs
Validation	5-6	Simulation execution; automated evaluation; analysis	Evaluation matrices; charts; statistical analysis	Completed simulations; evaluator consistency >0.8
Analysis	7	Results interpretation; visualization; limitation analysis	Results section; figures; Discussion	Clear differences; professional visuals; honest assessment
Documentation	8	PMT completion; appendix assembly; peer review	PMT v1.0; appendices; peer feedback	Acceptance criteria met; reproducible artifacts
Finalization	9	Revision integration; QA; thesis planning	Final PMT v2.0; thesis outline; presentation	Polished manuscript; clear roadmap

Activity Details:

Weeks 1-2 (Foundation):

- Systematic literature review (personality-aware dialogue, psychological frameworks, evaluation methodologies)

- RQ refinement with supervisors; scope validation
- N8N and Docker environment setup
- LLM API access verification and quota establishment
- Initial workflow prototyping and feasibility testing

Weeks 3-4 (Implementation):

- N8N workflow nodes implementation with error handling
- API integration for personality detection and response generation
- Prompt engineering: detection (JSON schema), generation (directive-driven), evaluation (rubric-based)
- A/B testing of prompt variations
- Comprehensive logging system and automated data export
- Version control setup and testing framework

Weeks 5-6 (Validation):

- Type A and Type B simulation execution (90-135 conversations)
- Multi-instance data collection for statistical reliability
- Real-time performance monitoring and QA checks
- LLM evaluator deployment with bias controls (randomization, blinding)
- Descriptive statistics and comparative analysis
- Effect size calculations and significance testing

Week 7 (Analysis):

- Comprehensive statistical analysis of experimental outcomes
- Qualitative analysis of conversation examples
- Comparison with Devdas (2025) baseline results
- Publication-quality visualization (time-series plots, bar charts with 95% CI)
- Limitation and validity threat analysis

Week 8 (Documentation):

- PMT manuscript integration and coherence review
- Comprehensive appendix assembly (prompts, schemas, analysis code)
- APA reference formatting and validation
- Internal QA and peer review submission

Week 9 (Finalization):

- Integration of supervisor and peer reviewer feedback
- Final proofreading and formatting consistency checks

- Thesis proposal outline development
- Ethical approval planning for thesis phase
- Conference-style presentation preparation

Resource Requirements:

Technical Resources:

- N8N Pro license (if needed for advanced features)
- LLM API credits: ~$200-500 (GPT-4, fallback to Claude)
- Cloud computing resources for backup and collaboration
- R/Python with statistical libraries (matplotlib, seaborn, scipy)

Human Resources:

- Supervisor meetings: 2 hours weekly for guidance and feedback
- Peer reviewers: 2-3 qualified reviewers for manuscript assessment
- Psychology domain expert: Personality profile validation
- Technical support: N8N and API troubleshooting as needed

Institutional Support:

- Access to academic databases (ACM, IEEE, PsycINFO)
- Computational resources for data analysis and visualization
- Presentation opportunities for feedback and validation
- Ethics board consultation for future human studies

6.3 Risk Management

Risk Assessment Framework:

Likelihood and Impact are assessed on a 3-level scale (Low/Medium/High). Risks with Medium×High or High×Medium scores trigger mitigation playbooks and weekly review.

Table 9. Project Risk Matrix and Mitigation Strategies

Risk Category	Specific Risk	Likelihood	Impact	Mitigation Strategy	Early Indicators
Data Quality	Simulated profiles lack realism	Medium	Medium	Psychology expert validation; mid-range profiles; pilot human validation	Low evaluator agreement; anomalous patterns
Evaluation	LLM evaluator	Medium	High	Fixed prompts (hashed); 3×	Consistency < 0.85; score

Risk Category	Specific Risk	Likelihood	Impact	Mitigation Strategy	Early Indicators
	bias/drift			runs; randomized order; human spot-checks	shifts
Technical	Prompt sensitivity	Medium	Medium	Prompt versioning; A/B tests; directive auditing	JSON regression; style oscillations
Reproducibility	Non-determinism	Medium	High	Pin model versions; fix seeds; config logging	Re-runs diverge > 5%
Performance	Latency exceeding thresholds	Low	Medium	Timeout caps; token budgets	p95 latency > 2s; timeouts > 1%
Ethics/Privacy	Data protection gaps	Low	High	Anonymization; encryption; access control; IRB	Missing logs; unauthorized access
Infrastructure	Vendor interruptions	Medium	Medium	Provider abstraction; alternates; backoff/retry	API errors; throttling
Scope	Feature creep	Medium	Medium	Scope guardrails; change control; defer to thesis	Backlog growth; missed milestones

Mitigation Strategies:

Data Quality Assurance:

- Validate simulated personality profiles with psychology domain experts
- Supplement extreme profiles (Type A/B) with mid-range combinations (Type C)
- Pilot human validation in thesis phase to assess ecological validity

Evaluation Reliability:

- Fixed evaluation prompts with cryptographic hashing to prevent drift
- Multiple independent evaluation runs (3×) with consistency analysis
- Randomized assessment order to prevent systematic biases

- Human spot-checking of automated evaluations (10% sample)
- Inter-rater reliability testing in thesis phase

Technical Robustness:

- Systematic prompt versioning with semantic version numbers (v1.0.0)
- A/B testing of prompt variations to identify optimal formulations
- Comprehensive auditing of generated directives
- Maintenance of prompt libraries with detailed change logs

Reproducibility Assurance:

- Pin specific model versions with documented API endpoints
- Fix random seeds for all stochastic processes
- Archive complete configuration snapshots for each experimental run
- Comprehensive logging of all system interactions
- Automated reproduction scripts for key findings

Mitigation Playbooks (triggered by early indicators):

- **Evaluator drift (consistency < 0.85):** Freeze current prompts; rerun with fixed seeds; compare across providers (GPT-4, Claude); re-baseline with human spot-checks; document deltas
- **JSON contract failures (>0.5%):** Enable strict schema validation with auto-retry; reduce temperature; add minimal repair step; log offending examples for prompt hardening
- **Latency regressions (p95 > 2s):** Reduce max_tokens; cache static response sections; parallelize I/O operations; switch to alternate endpoint; monitor token usage patterns
- **Privacy incident:** Disable non-essential access immediately; rotate API keys; enable detailed audit logging; perform post-mortem analysis; add compensating controls

Contingency Planning:

Alternative Model Providers:

- Ready access to OpenAI GPT-4, Anthropic Claude with equivalent API interfaces
- All prompts and evaluation procedures designed to be provider-agnostic
- Rapid switching capability without experimental disruption

Evaluation Fallback Procedures:

- Human evaluation with trained raters (n=30) if automated evaluation proves unreliable

- Evaluation rubrics interpretable by both automated and human assessors
- Relationships with qualified evaluators for rapid deployment

Scope Adjustment Gates:

- **Evaluator consistency < 0.75 after two fixes:** Switch to human evaluation sample; freeze prompt changes
- **JSON compliance < 98.5% for two weeks:** Prioritize contract hardening over feature work
- **Latency p95 > 3s in week 5:** Drop non-critical nodes; defer features to thesis phase

Timeline Buffer Management:

- Built-in buffers for unexpected challenges
- Non-critical features identified for potential deferral to thesis phase
- Regular milestone reviews for early identification of delays

Monitoring and Review Procedures:

Weekly Risk Assessment:

- Progress against planned milestones
- Emergence of new risks or changes in existing risk profiles
- Effectiveness of current mitigation strategies
- Need for additional resources or support

Stakeholder Communication:

- Regular updates to supervisors with explicit risk status reporting
- Early escalation of issues requiring additional resources or guidance
- Monthly progress reports with updated risk assessments

Operational KPIs (tracked weekly):

KPI	Target	Action Threshold
JSON contract compliance	≥ 99.5%	< 98.5% triggers contract hardening playbook
EMA stabilization by turn 8	≥ 80% sessions	< 70% triggers detection prompt review
Evaluator inter-run consistency	≥ 0.85	< 0.80 triggers drift investigation
p95 latency	< 2.0s	> 2.5s triggers performance playbook
Incident rate (per 1k turns)	< 1.0	≥ 2.0 triggers post-mortem and fixes
Privacy/security incidents	0	Any incident triggers escalation and audit

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8. Appendix

Note: This appendix section provides technical specifications, evaluation materials, configuration details, and a glossary of key terms to support reproducibility and accessibility for interdisciplinary readers.

Appendix A. Prompt Interfaces and Templates

A.1 Detection Prompt (JSON-Structured Response)

```
You are a Zurich Model personality and stress assessor. Analyze the last user
message and up to 10 prior turns.

Return ONLY valid JSON in this exact format:
{
  "ocean": {"O": -1.0 to 1.0, "C": -1.0 to 1.0, "E": -1.0 to 1.0, "A": -1.0
to 1.0, "N": -1.0 to 1.0},
  "confidence": {"O": 0.0-1.0, "C": 0.0-1.0, "E": 0.0-1.0, "A": 0.0-1.0, "N":
0.0-1.0},
  "stress": {
    "level": 0|1|2|3|4,
    "drivers":
["workload"|"uncertainty"|"social"|"health"|"finance"|"other"...],
    "signals": {"cognitive":[], "affective":[], "physiological":[],
"behavioral":[]},
    "confidence": 0.0-1.0
  },
  "coaching_mode": "vent_validate"|"plan_structure"|"cope_rehearse",
  "evidence_quotes": [string, ...]
}

Rules:
- Base all inferences strictly on quoted spans from the user's text; no
external facts.
- Stress level: 0 none, 1 mild, 2 moderate, 3 high, 4 crisis.
- Choose coaching_mode that best fits immediate need: vent_validate
(validation), plan_structure (planning), cope_rehearse (immediate coping).

Conversation: {conversation_context}
User message: {user_message}
```

A.2 Generation Prompt with Behavioral Directives

```
You are a supportive conversational assistant. Follow these behavioral
directives strictly:
{behavioral_directives}

CRITICAL CONSTRAINTS:
```

```
1. Ground all responses strictly in the user's text - no external information
or assumptions
2. Respond in 70-150 words
3. Ask at most 1-2 questions
4. Maintain supportive, empathetic tone appropriate to the directives
5. Coaching mode: {coaching_mode}
6. If stress_level ≥ 3 and crisis indicators appear, provide a neutral
supportive message and suggest reaching out to a trusted person or local
support line; do not provide clinical advice.

User context: "{conversation_context}"
User message: "{user_message}"
```

A.3 Evaluation Prompt for Automated Assessment

Used for LLM-based automated evaluation of chatbot responses (Section 4.8, Verification Node)

```
You are an expert evaluator assessing personality-aware chatbot responses.
Evaluate this chatbot response using the specified criteria. Provide only the
requested assessment format with no additional commentary.

CONVERSATION CONTEXT:
User: "{user_message}"
Assistant: "{assistant_response}"
Detected Personality: {ocean_disc}
Applied Directives: {directives}

EVALUATION CRITERIA:

1. Detection Accuracy: Does the detected personality profile (ocean_disc)
appropriately reflect observable personality cues in the user's message?
Consider linguistic markers, emotional expression, and behavioral patterns.

2. Regulation Effectiveness: Are the specified behavioral directives
correctly and consistently applied throughout the assistant's response? Check
for adherence to all listed directives.

3. Emotional Tone Appropriate: Does the response's emotional tone match both
the user's current emotional state AND their personality profile? Consider
warmth, formality, energy level, and empathy.

4. Relevance & Coherence: Is the response contextually relevant to the user's
message and logically coherent? Does it directly address the user's concerns
without tangential content?

5. Personality Needs Addressed: Does the response effectively address trait-
specific emotional and interactional needs? For example:
   - High N: Reassurance and validation
```

```
  - Low N: Pragmatic, solution-focused
  - High E: Energetic, collaborative framing
  - Low E: Calm, reflective tone
  - High O: Novel ideas and creative perspectives
  - Low O: Concrete, practical examples
  - High C: Structured plans with clear steps
  - Low C: Flexible, adaptable guidance
  - High A: Warm, cooperative language
  - Low A: Direct, matter-of-fact stance

RESPONSE GUIDELINES:
For each criterion, respond with ONLY one of these three values:
- "Yes" = Clear success, criterion fully met with strong evidence
- "Partial" = Mixed performance, criterion partially met or unclear evidence
- "No" = Clear failure, criterion not met or contradicted

OUTPUT FORMAT (use this exact format, no additional text):
Detection_Accuracy: [Yes/Partial/No], Regulation_Effectiveness:
[Yes/Partial/No], Emotional_Tone: [Yes/Partial/No], Relevance_Coherence:
[Yes/Partial/No], Personality_Needs: [Yes/Partial/No]
```

Example Evaluation Output:

```
Detection_Accuracy: Yes, Regulation_Effectiveness: Yes, Emotional_Tone:
Partial, Relevance_Coherence: Yes, Personality_Needs: Yes
```

Appendix B. Evaluation Rubric and Scoring Guidelines

Table B.1. Comprehensive Evaluation Criteria Definitions

Criterion	Definition	"Yes" (2 points)	"Partial" (1 point)	"No" (0 points)
Detection Accuracy	Alignment between `ocean_disc` inference and observable personality cues	Clear trait indicators match detected values; logical inference	Some alignment but missing or questionable trait assessments	Poor alignment or systematic misdetection
Regulation Effectiveness	Appropriate application of trait-specific behavioral strategies	All relevant directives correctly implemented in response style	Partial directive application or minor inconsistencies	Directives ignored or incorrectly applied
Emotional Tone Appropriate	Match between response emotional tone and user	Tone perfectly suited to user's emotional state and	Generally appropriate tone with minor misalignments	Tone inappropriate or potentially harmful

Criterion	Definition	"Yes" (2 points)	"Partial" (1 point)	"No" (0 points)
	state/personality	personality traits		
Relevance & Coherence	Contextual appropriateness and logical consistency of response	Highly relevant, coherent, and well-structured response	Generally relevant with minor coherence issues	Off-topic, incoherent, or poorly structured
Personality Needs Addressed	Satisfaction of trait-specific emotional and interactional requirements	Response clearly addresses personality-driven needs	Partially addresses needs or generic support	Fails to address personality-specific needs

B.2 Scoring Aggregation Procedures

- **Row-level scoring:** Sum of all criterion scores for single interaction
- **Bot-level scoring:** Average of all row scores for individual bot instance
- **Condition-level scoring:** Average across all bots in regulated or baseline condition
- **Statistical analysis:** T-tests, effect sizes, confidence intervals for group comparisons

Appendix C. Technical Configurations and System Specifications

C.1 Model Specifications and Parameters

```yaml
# LLM Configuration
detection_model:
  name: "gemini-1.5-pro"
  endpoint: "https://ai.juguang.chat/v1/chat/completions"
  parameters:
    temperature: 0.1
    max_tokens: 200
    timeout_seconds: 20
    top_p: 1.0

generation_model:
  name: "gemini-1.5-pro"
  endpoint: "https://ai.juguang.chat/v1/chat/completions"
  parameters:
    temperature: 0.7
    max_tokens: 220
    timeout_seconds: 20
    top_p: 1.0

# Alternative Configuration
```

```yaml
fallback_model:
  name: "gpt-4-turbo"
  endpoint: "https://api.openai.com/v1/chat/completions"
  parameters:
    temperature: 0.1  # detection
    # temperature: 0.7  # generation
    max_tokens: 200
    timeout_seconds: 20
```

C.2 N8N Workflow Node Configuration

Note: Complete N8N workflow JSON available in repository: `preliminary-studies/w9-Technical-Specifications/workflows/webhook-personality-pipeline.json`

Example: Personality Detection Node (JavaScript)

```javascript
// File: n8n-nodes/detect-ocean.js
// Node: Zurich Model Detection (GPT-4 + EMA)
// Purpose: Continuous OCEAN inference with confidence-weighted EMA smoothing

const EMA_ALPHA = 0.3;  // Smoothing factor
const MIN_CONFIDENCE = 0.4;  // Minimum confidence for updates
const STABILIZATION_TURNS = 6;  // Turns required for stable designation

// Extract inputs from N8N workflow
const sessionId = $input.item.json.session_id;
const userMessage = $input.item.json.message;
const conversationHistory = $input.item.json.conversation_context || [];
const previousState = $input.item.json.previous_state || null;

// Build prompt for LLM
const detectionPrompt = `You are a Zurich Model personality assessor. Analyze
the user's message and conversation history to infer OCEAN personality
traits.

Conversation History:
${conversationHistory.map(turn => `User: ${turn.user}\nAssistant:
${turn.assistant}`).join('\n\n')}

Current User Message: "${userMessage}"

Return ONLY valid JSON in this exact format:
{
  "ocean": {"O": <-1.0 to 1.0>, "C": <-1.0 to 1.0>, "E": <-1.0 to 1.0>, "A":
<-1.0 to 1.0>, "N": <-1.0 to 1.0>},
  "confidence": {"O": <0.0 to 1.0>, "C": <0.0 to 1.0>, "E": <0.0 to 1.0>,
"A": <0.0 to 1.0>, "N": <0.0 to 1.0>},
  "reasoning": "Brief explanation of detected motivations"
}`;
```

```
// Call LLM API (OpenAI-compatible endpoint)
const llmResponse = await
fetch('https://ai.juguang.chat/v1/chat/completions', {
  method: 'POST',
  headers: {
    'Content-Type': 'application/json',
    'Authorization': `Bearer ${process.env.JUGUANG_API_KEY}`
  },
  body: JSON.stringify({
    model: 'gemini-1.5-pro',
    messages: [
      {role: 'system', content: 'You are a personality assessment expert.'},
      {role: 'user', content: detectionPrompt}
    ],
    temperature: 0.1,
    max_tokens: 200
  })
});

const llmData = await llmResponse.json();
const rawDetection = JSON.parse(llmData.choices[0].message.content);

// Apply EMA smoothing if previous state exists
let smoothedOcean = {};
let smoothedConfidence = {};

const traits = ['O', 'C', 'E', 'A', 'N'];
for (const trait of traits) {
  const currentValue = rawDetection.ocean[trait];
  const currentConf = rawDetection.confidence[trait];

  if (previousState && currentConf >= MIN_CONFIDENCE) {
    // EMA: smoothed = α * current + (1 - α) * previous
    smoothedOcean[trait] = EMA_ALPHA * currentValue + (1 - EMA_ALPHA) *
previousState.ocean[trait];
    smoothedConfidence[trait] = Math.max(currentConf,
previousState.confidence[trait]);
  } else if (previousState) {
    // Low confidence: keep previous value
    smoothedOcean[trait] = previousState.ocean[trait];
    smoothedConfidence[trait] = previousState.confidence[trait];
  } else {
    // First turn: use raw detection
    smoothedOcean[trait] = currentValue;
    smoothedConfidence[trait] = currentConf;
  }
}

// Determine stability (requires 6+ consistent turns with low variance)
```

```
const turnCount = (previousState?.turn_count || 0) + 1;
const variance = previousState ? calculateVariance(smoothedOcean,
previousState.ocean) : 1.0;
const stable = (turnCount >= STABILIZATION_TURNS && variance < 0.15);

// Helper function to calculate trait variance
function calculateVariance(current, previous) {
  const diffs = traits.map(t => Math.abs(current[t] - previous[t]));
  return diffs.reduce((a, b) => a + b, 0) / diffs.length;
}

// Return output for next N8N node
return {
  json: {
    session_id: sessionId,
    ocean: smoothedOcean,
    confidence: smoothedConfidence,
    stable: stable,
    turn_count: turnCount,
    variance: variance,
    reasoning: rawDetection.reasoning
  }
};
```

Example: Regulation Node (Directive Mapping)

```
// File: n8n-nodes/enhanced-regulation.js
// Node: Enhanced Regulation
// Purpose: Map continuous OCEAN traits to behavioral directives

const DIRECTIVE_THRESHOLD = 0.2;  // Minimum |trait| value for activation

// Extract personality state
const ocean = $input.item.json.ocean;
const confidence = $input.item.json.confidence;
const stable = $input.item.json.stable;

// Initialize directive list
const directives = [];

// Openness directives (Arousal domain)
if (ocean.O > DIRECTIVE_THRESHOLD && confidence.O >= 0.4) {
  const intensity = ocean.O;
  if (intensity > 0.6) {
    directives.push("Strongly emphasize novel concepts and creative
perspectives");
  } else {
    directives.push("Introduce some novel ideas and fresh approaches");
  }
```

```
} else if (ocean.O < -DIRECTIVE_THRESHOLD && confidence.O >= 0.4) {
  const intensity = Math.abs(ocean.O);
  if (intensity > 0.6) {
    directives.push("Use concrete, familiar examples; avoid abstract
novelty");
  } else {
    directives.push("Prefer practical, proven approaches");
  }
}

// Conscientiousness directives (Structure domain)
if (ocean.C > DIRECTIVE_THRESHOLD && confidence.C >= 0.4) {
  const intensity = ocean.C;
  if (intensity > 0.6) {
    directives.push("Provide detailed structured plans with clear steps,
timelines, and organization");
  } else {
    directives.push("Offer structured guidance with clear next steps");
  }
} else if (ocean.C < -DIRECTIVE_THRESHOLD && confidence.C >= 0.4) {
  const intensity = Math.abs(ocean.C);
  if (intensity > 0.6) {
    directives.push("Strongly emphasize flexibility; actively avoid rigid
frameworks");
  } else {
    directives.push("Keep guidance flexible; minimize prescriptive
structure");
  }
}

// Extraversion directives (Arousal domain)
if (ocean.E > DIRECTIVE_THRESHOLD && confidence.E >= 0.4) {
  directives.push("Use energetic, engaging tone; frame as collaborative");
} else if (ocean.E < -DIRECTIVE_THRESHOLD && confidence.E >= 0.4) {
  directives.push("Adopt calm, reflective tone; respect need for
introspection");
}

// Agreeableness directives (Affiliation domain)
if (ocean.A > DIRECTIVE_THRESHOLD && confidence.A >= 0.4) {
  directives.push("Use warm, empathetic language; emphasize cooperation");
} else if (ocean.A < -DIRECTIVE_THRESHOLD && confidence.A >= 0.4) {
  directives.push("Maintain matter-of-fact, direct stance; respect
autonomy");
}

// Neuroticism directives (Security domain)
if (ocean.N > DIRECTIVE_THRESHOLD && confidence.N >= 0.4) {
  const intensity = ocean.N;
```

```
  if (intensity > 0.6) {
    directives.push("Provide strong reassurance; acknowledge concerns; offer
detailed coping strategies");
  } else {
    directives.push("Provide gentle reassurance and emotional validation");
  }
} else if (ocean.N < -DIRECTIVE_THRESHOLD && confidence.N >= 0.4) {
  directives.push("Match confident tone; focus on pragmatic solutions");
}

// Fallback if no directives triggered
if (directives.length === 0) {
  directives.push("Use balanced, neutral tone appropriate for general
support");
}

return {
  json: {
    session_id: $input.item.json.session_id,
    directives: directives,
    ocean: ocean,
    confidence: confidence,
    stable: stable
  }
};
```

C.3 Environment Configuration

```
# .env file template
N8N_BASIC_AUTH_ACTIVE=true
N8N_BASIC_AUTH_USER=research_user
N8N_BASIC_AUTH_PASSWORD=secure_password_here

GEMINI_API_KEY=your_gemini_api_key_here
OPENAI_API_KEY=your_openai_api_key_here
JUGUANG_API_KEY=your_juguang_api_key_here

# Logging configuration
LOG_LEVEL=info
LOG_FILE_PATH=./logs/personality_chatbot.log
EXPORT_FORMAT=jsonl

# Database configuration (for extended version)
POSTGRES_HOST=localhost
POSTGRES_PORT=5432
POSTGRES_DB=personality_research
POSTGRES_USER=research_user
POSTGRES_PASSWORD=secure_db_password
```

Appendix D. Statistical Analysis and Visualization Code

D.1 Data Analysis Pipeline

Prerequisites: Python 3.8+, pip install pandas numpy matplotlib seaborn scipy

Expected CSV Structure:

```
# File: evaluation_results.csv
# Columns required:
# - Row (int): Row number
# - Condition (str): "Regulated" or "Baseline"
# - Personality_Type (str): "Type_A", "Type_B", or "Type_C"
# - Turn (int): Conversation turn number
# - User_Message (str): User input text
# - Assistant_Reply (str): System response
# - Detection_Accuracy (str): "Yes", "Partial", or "No"
# - Regulation_Effectiveness (str): "Yes", "Partial", or "No" (Regulated
only)
# - Emotional_Tone (str): "Yes", "Partial", or "No"
# - Relevance_Coherence (str): "Yes", "Partial", or "No"
# - Personality_Needs (str): "Yes", "Partial", or "No"
# - Total_Score (int): Sum of all criterion scores
```

Complete Analysis Script:

```python
#!/usr/bin/env python3
"""
Personality-Aware Chatbot Evaluation Analysis Pipeline
Author: [Your Name]
Date: 2024-10-15
Purpose: Analyze evaluation results from personality-adaptive vs. baseline
chatbot conditions
"""

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from pathlib import Path

# Configuration
EVALUATION_FILE = 'evaluation_results.csv'  # Input CSV path
OUTPUT_DIR = 'analysis_output'  # Directory for outputs
ALPHA = 0.05  # Significance level
RANDOM_SEED = 42  # For reproducibility

# Set random seed for reproducibility
np.random.seed(RANDOM_SEED)
```

```python
# Configure plotting style
sns.set_style('whitegrid')
sns.set_palette('husl')
plt.rcParams['figure.dpi'] = 300
plt.rcParams['font.size'] = 10

def setup_output_directory():
    """Create output directory if it doesn't exist"""
    Path(OUTPUT_DIR).mkdir(parents=True, exist_ok=True)
    print(f"✓ Output directory created: {OUTPUT_DIR}/")

def load_evaluation_data(filepath):
    """
    Load and preprocess evaluation results from CSV

    Args:
        filepath (str): Path to evaluation CSV file

    Returns:
        pd.DataFrame: Preprocessed evaluation data

    Raises:
        FileNotFoundError: If CSV file doesn't exist
        ValueError: If required columns are missing
    """
    try:
        df = pd.read_csv(filepath)
        print(f"✓ Loaded {len(df)} rows from {filepath}")
    except FileNotFoundError:
        raise FileNotFoundError(
            f"Evaluation file not found: {filepath}\n"
            f"Please ensure CSV file exists with required columns."
        )

    # Validate required columns
    required_cols = ['Condition', 'Personality_Type', 'Detection_Accuracy',
                     'Emotional_Tone', 'Relevance_Coherence',
'Personality_Needs']
    missing_cols = [col for col in required_cols if col not in df.columns]
    if missing_cols:
        raise ValueError(f"Missing required columns: {missing_cols}")

    # Convert scoring to numeric
    score_mapping = {'Yes': 2, 'Partial': 1, 'No': 0}
    score_columns = ['Detection_Accuracy', 'Regulation_Effectiveness',
                     'Emotional_Tone', 'Relevance_Coherence',
'Personality_Needs']
```

```
    for col in score_columns:
        if col in df.columns:
            original_count = df[col].notna().sum()
            df[col] = df[col].map(score_mapping)
            converted_count = df[col].notna().sum()
            if converted_count < original_count:
                print(f"⚠ Warning: {col} had {original_count -
converted_count} unmapped values")

    # Calculate total score if not present
    if 'Total_Score' not in df.columns:
        score_cols_present = [c for c in score_columns if c in df.columns]
        df['Total_Score'] = df[score_cols_present].sum(axis=1)
        print(f"✓ Calculated Total_Score from {len(score_cols_present)}
criteria")

    print(f"✓ Data preprocessing complete: {len(df)} valid rows")
    return df

def analyze_results(df):
    """
    Perform statistical analysis comparing regulated vs baseline conditions

    Args:
        df (pd.DataFrame): Preprocessed evaluation data

    Returns:
        dict: Statistical results for each criterion including means, t-
tests, effect sizes
    """
    regulated = df[df['Condition'] == 'Regulated']
    baseline = df[df['Condition'] == 'Baseline']

    print(f"\n📊 Statistical Analysis:")
    print(f"  Regulated condition: {len(regulated)} observations")
    print(f"  Baseline condition: {len(baseline)} observations")

    # Shared criteria comparison (both conditions have these)
    shared_criteria = ['Emotional_Tone', 'Relevance_Coherence',
'Personality_Needs']

    results = {}
    for criterion in shared_criteria:
        reg_scores = regulated[criterion].dropna()
        base_scores = baseline[criterion].dropna()

        # Independent samples t-test
```

```
        t_stat, p_value = stats.ttest_ind(reg_scores, base_scores)

        # Effect size (Cohen's d)
        # d = (M1 - M2) / pooled_std
        pooled_std = np.sqrt(((len(reg_scores) - 1) * reg_scores.var() +
                              (len(base_scores) - 1) * base_scores.var()) /
                             (len(reg_scores) + len(base_scores) - 2))
        cohens_d = (reg_scores.mean() - base_scores.mean()) / pooled_std if
pooled_std > 0 else 0

        # 95% Confidence Interval for mean difference
        se_diff = pooled_std * np.sqrt(1/len(reg_scores) +
1/len(base_scores))
        df_total = len(reg_scores) + len(base_scores) - 2
        t_critical = stats.t.ppf(0.975, df_total)  # 95% CI
        ci_lower = (reg_scores.mean() - base_scores.mean()) - t_critical *
se_diff
        ci_upper = (reg_scores.mean() - base_scores.mean()) + t_critical *
se_diff

        results[criterion] = {
            'regulated_mean': reg_scores.mean(),
            'regulated_std': reg_scores.std(),
            'baseline_mean': base_scores.mean(),
            'baseline_std': base_scores.std(),
            'difference': reg_scores.mean() - base_scores.mean(),
            'ci_95_lower': ci_lower,
            'ci_95_upper': ci_upper,
            't_statistic': t_stat,
            'p_value': p_value,
            'cohens_d': cohens_d,
            'significant': p_value < ALPHA,
            'n_regulated': len(reg_scores),
            'n_baseline': len(base_scores)
        }

        # Print summary
        sig_marker = "***" if p_value < 0.001 else "**" if p_value < 0.01
else "*" if p_value < 0.05 else "ns"
        print(f"\n  {criterion}:")
        print(f"    Regulated: M={reg_scores.mean():.2f},
SD={reg_scores.std():.2f}")
        print(f"    Baseline:  M={base_scores.mean():.2f},
SD={base_scores.std():.2f}")
        print(f"    t({df_total})={t_stat:.2f}, p={p_value:.4f} {sig_marker},
d={cohens_d:.2f}")
        print(f"    95% CI: [{ci_lower:.2f}, {ci_upper:.2f}]")

    return results
```

```
def create_comparison_plot(df, save_path=None):
    """
    Create publication-quality comparison visualization

    Args:
        df (pd.DataFrame): Evaluation data
        save_path (str, optional): Path to save figure. If None, displays
interactively.
    """
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 6))

    # Plot 1: Overall performance by condition and personality type
    # Group by condition and personality type, calculate mean total score
    performance_data = df.groupby(['Condition',
'Personality_Type'])['Total_Score'].mean().unstack()
    performance_data.plot(kind='bar', ax=ax1, color=['#3498db', '#e74c3c',
'#2ecc71'], alpha=0.8)
    ax1.set_title('Overall Performance by Condition and Personality Type',
fontsize=12, fontweight='bold')
    ax1.set_ylabel('Mean Total Score', fontsize=11)
    ax1.set_xlabel('Condition', fontsize=11)
    ax1.legend(title='Personality Type', fontsize=9)
    ax1.grid(axis='y', alpha=0.3)
    ax1.set_xticklabels(ax1.get_xticklabels(), rotation=0)

    # Plot 2: Criterion-wise comparison (shared criteria only)
    criteria = ['Emotional_Tone', 'Relevance_Coherence', 'Personality_Needs']
    criterion_labels = ['Emotional\nTone', 'Relevance &\nCoherence',
'Personality\nNeeds']

    regulated_means = [df[df['Condition'] == 'Regulated'][col].mean() for col
in criteria]
    baseline_means = [df[df['Condition'] == 'Baseline'][col].mean() for col
in criteria]

    # Calculate standard errors for error bars
    regulated_se = [df[df['Condition'] == 'Regulated'][col].sem() for col in
criteria]
    baseline_se = [df[df['Condition'] == 'Baseline'][col].sem() for col in
criteria]

    x = np.arange(len(criteria))
    width = 0.35

    ax2.bar(x - width/2, regulated_means, width, label='Regulated',
            color='#3498db', alpha=0.8, yerr=regulated_se, capsize=5)
    ax2.bar(x + width/2, baseline_means, width, label='Baseline',
            color='#e74c3c', alpha=0.8, yerr=baseline_se, capsize=5)
```

```
    ax2.set_title('Criterion-wise Performance Comparison', fontsize=12,
fontweight='bold')
    ax2.set_ylabel('Mean Score (0-2 scale)', fontsize=11)
    ax2.set_xlabel('Evaluation Criterion', fontsize=11)
    ax2.set_xticks(x)
    ax2.set_xticklabels(criterion_labels, rotation=0)
    ax2.legend(fontsize=10, loc='upper left')
    ax2.grid(axis='y', alpha=0.3)
    ax2.set_ylim([0, 2])

    plt.tight_layout()

    if save_path:
        plt.savefig(save_path, dpi=300, bbox_inches='tight')
        print(f"✓ Figure saved to: {save_path}")
    else:
        plt.show()

def generate_summary_report(results, output_path='analysis_summary.txt'):
    """
    Generate human-readable summary report of statistical analysis

    Args:
        results (dict): Statistical results from analyze_results()
        output_path (str): Path to save summary report
    """
    with open(output_path, 'w') as f:
        f.write("=" * 70 + "\n")
        f.write("PERSONALITY-AWARE CHATBOT EVALUATION SUMMARY\n")
        f.write("=" * 70 + "\n\n")

        for criterion, stats in results.items():
            f.write(f"\n{criterion.replace('_', ' ').upper()}\n")
            f.write("-" * 70 + "\n")
            f.write(f"  Regulated Condition:  M =
{stats['regulated_mean']:.3f}, SD = {stats['regulated_std']:.3f} (n =
{stats['n_regulated']})\n")
            f.write(f"  Baseline Condition:   M =
{stats['baseline_mean']:.3f}, SD = {stats['baseline_std']:.3f} (n =
{stats['n_baseline']})\n")
            f.write(f"  Mean Difference:      {stats['difference']:.3f}\n")
            f.write(f"  95% CI:               [{stats['ci_95_lower']:.3f},
{stats['ci_95_upper']:.3f}]\n")
            f.write(f"  t-statistic:          {stats['t_statistic']:.3f}\n")
            f.write(f"  p-value:              {stats['p_value']:.4f} {'***'
if stats['p_value'] < 0.001 else '**' if stats['p_value'] < 0.01 else '*' if
stats['p_value'] < 0.05 else 'ns'}\n")
            f.write(f"  Cohen's d:            {stats['cohens_d']:.3f}
```

```
({'large' if abs(stats['cohens_d']) >= 0.8 else 'medium' if
abs(stats['cohens_d']) >= 0.5 else 'small'})\n")
            f.write(f"  Statistically Significant: {'Yes' if
stats['significant'] else 'No'} (α = {ALPHA})\n")

        f.write("\n" + "=" * 70 + "\n")
        f.write("Note: *** p < 0.001, ** p < 0.01, * p < 0.05, ns = not
significant\n")
        f.write("Cohen's d interpretation: small (0.2), medium (0.5), large
(0.8)\n")
        f.write("=" * 70 + "\n")

    print(f"✓ Summary report saved to: {output_path}")

def main():
    """
    Main execution function for complete analysis pipeline

    Usage:
        python analysis_pipeline.py

    Outputs:
        - analysis_output/performance_comparison.png (visualization)
        - analysis_output/analysis_summary.txt (statistical summary)
        - analysis_output/results_detailed.csv (detailed results)
    """
    print("=" * 70)
    print("PERSONALITY-AWARE CHATBOT EVALUATION ANALYSIS PIPELINE")
    print("=" * 70)

    # Step 1: Setup
    setup_output_directory()

    # Step 2: Load data
    print(f"\n📂 Loading evaluation data from: {EVALUATION_FILE}")
    try:
        df = load_evaluation_data(EVALUATION_FILE)
    except (FileNotFoundError, ValueError) as e:
        print(f"❌ Error: {e}")
        print("\n💡 Expected CSV format:")
        print("
Condition,Personality_Type,Turn,User_Message,Assistant_Reply,")
        print("
Detection_Accuracy,Regulation_Effectiveness,Emotional_Tone,")
        print("   Relevance_Coherence,Personality_Needs,Total_Score")
        return

    # Step 3: Analyze
    results = analyze_results(df)
```

```
    # Step 4: Generate visualizations
    print(f"\n📈 Generating visualizations...")
    create_comparison_plot(df,
save_path=f'{OUTPUT_DIR}/performance_comparison.png')

    # Step 5: Generate summary report
    print(f"\n📄 Generating summary report...")
    generate_summary_report(results,
output_path=f'{OUTPUT_DIR}/analysis_summary.txt')

    # Step 6: Save detailed results
    results_df = pd.DataFrame(results).T
    results_df.to_csv(f'{OUTPUT_DIR}/results_detailed.csv',
index_label='Criterion')
    print(f"✓ Detailed results saved to: {OUTPUT_DIR}/results_detailed.csv")

    print(f"\n✅ Analysis complete! All outputs saved to: {OUTPUT_DIR}/")
    print("=" * 70)

# Entry point
if __name__ == "__main__":
    main()
```

D.2 Example Data Generation Script

For testing the analysis pipeline without real evaluation data:

```
#!/usr/bin/env python3
"""
Generate synthetic evaluation data for testing the analysis pipeline
"""

import pandas as pd
import numpy as np

np.random.seed(42)

def generate_synthetic_data(n_rows=180):
    """
    Generate synthetic evaluation data matching expected format

    Args:
        n_rows (int): Number of rows to generate (must be divisible by 6 for
balanced design)

    Returns:
        pd.DataFrame: Synthetic evaluation data
    """
```

```
    conditions = ['Regulated', 'Baseline']
    personality_types = ['Type_A', 'Type_B', 'Type_C']

    # Regulated condition has better performance (+0.4 mean difference)
    regulated_boost = 0.4

    data = []
    row_id = 0

    for condition in conditions:
        for personality_type in personality_types:
            for turn in range(1, int(n_rows / 6) + 1):
                # Base scores with some personality-type variation
                base_mean = 1.2 if personality_type == 'Type_A' else 1.0 if
personality_type == 'Type_B' else 0.8

                # Regulated condition gets boost
                if condition == 'Regulated':
                    base_mean += regulated_boost

                # Generate scores with realistic noise
                scores = {}
                for criterion in ['Detection_Accuracy', 'Emotional_Tone',
'Relevance_Coherence', 'Personality_Needs']:
                    score = base_mean + np.random.normal(0, 0.3)
                    score = max(0, min(2, score))  # Clip to [0, 2]
                    # Convert to categorical
                    if score >= 1.5:
                        scores[criterion] = 'Yes'
                    elif score >= 0.5:
                        scores[criterion] = 'Partial'
                    else:
                        scores[criterion] = 'No'

                # Regulation effectiveness only for regulated condition
                if condition == 'Regulated':
                    reg_eff = base_mean + np.random.normal(0.2, 0.3)
                    reg_eff = max(0, min(2, reg_eff))
                    scores['Regulation_Effectiveness'] = 'Yes' if reg_eff >=
1.5 else 'Partial' if reg_eff >= 0.5 else 'No'
                else:
                    scores['Regulation_Effectiveness'] = None

                # Calculate total score (numeric)
                score_map = {'Yes': 2, 'Partial': 1, 'No': 0}
                total = sum(score_map.get(scores[c], 0) for c in
['Detection_Accuracy', 'Emotional_Tone', 'Relevance_Coherence',
'Personality_Needs'])
                if condition == 'Regulated' and
```

```
scores['Regulation_Effectiveness']:
                    total +=
score_map.get(scores['Regulation_Effectiveness'], 0)

                data.append({
                    'Row': row_id,
                    'Condition': condition,
                    'Personality_Type': personality_type,
                    'Turn': turn,
                    'User_Message': f"Sample user message for turn {turn}",
                    'Assistant_Reply': f"Sample assistant reply for
{condition} condition",
                    'Detection_Accuracy': scores['Detection_Accuracy'],
                    'Regulation_Effectiveness':
scores.get('Regulation_Effectiveness'),
                    'Emotional_Tone': scores['Emotional_Tone'],
                    'Relevance_Coherence': scores['Relevance_Coherence'],
                    'Personality_Needs': scores['Personality_Needs'],
                    'Total_Score': total
                })
                row_id += 1

    return pd.DataFrame(data)

if __name__ == "__main__":
    print("Generating synthetic evaluation data...")
    df = generate_synthetic_data(n_rows=180)
    df.to_csv('evaluation_results.csv', index=False)
    print(f"✓ Generated {len(df)} rows of synthetic data")
    print(f"✓ Saved to: evaluation_results.csv")
    print(f"\nData summary:")
    print(df.groupby('Condition')['Total_Score'].describe())
```

D.3 Reproducibility Checklist

- ☐ All model versions documented with API endpoints
- ☐ Random seeds fixed and recorded for all stochastic processes
- ☐ Configuration files archived with cryptographic hashes
- ☐ Complete interaction logs saved in JSONL format
- ☐ Analysis code version controlled with dependency specifications
- ☐ Environment setup documented with container configurations
- ☐ Evaluation procedures validated with consistency checks
- ☐ Statistical analysis methods justified and documented

This comprehensive appendix provides all necessary technical details for reproducing the experimental procedures, analyzing results, and extending the research framework. The

combination of detailed prompts, configuration specifications, and analysis code ensures that other researchers can replicate and build upon this work effectively.

Appendix E. Glossary of Key Terms

OCEAN: Acronym for the Big Five personality dimensions: Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism. Each trait is assessed on a continuous scale from -1.0 (low) to +1.0 (high).

EMA (Exponential Moving Average): A statistical smoothing technique that computes weighted averages giving exponentially decreasing weights to older observations. In this system, EMA stabilizes personality trait estimates across conversation turns by blending current detections (30% weight, $\alpha=0.3$) with historical averages (70% weight).

α (Alpha): The smoothing factor in EMA calculations, set to 0.3 in this implementation. Determines the balance between responsiveness to new observations (higher α) and stability from historical data (lower α).

JSON Contracts: Standardized data formats using JavaScript Object Notation (JSON) that define the exact structure of inputs and outputs for each system module, ensuring interoperability and enabling systematic testing.

Policy Packs: Modular configuration files (YAML/JSON format) that encapsulate domain-specific behavioral rules, coaching modes, and regulation directives. Enable rapid domain transfer without code modification.

JSONL (JSON Lines): A file format where each line is a valid JSON object, enabling efficient streaming and processing of structured logs. Used for comprehensive audit trails and experimental data export.

Zurich Model: A psychological framework conceptualizing human motivation through three fundamental systems: security (safety/comfort needs), arousal (novelty/stimulation needs), and affiliation (social connection needs). Used to map personality traits to behavioral adaptations.

Dialog Grounding: The constraint that all system responses must be strictly entailed by the user's conversational input, preventing fabrication of external information or facts not present in the dialogue.

Quote-and-Bound: A generation approach ensuring every assertion in responses is directly grounded in quoted user statements, critical for preventing hallucination in human-centered applications.

Confidence-Aware Filtering: A mechanism that only updates personality estimates when detection confidence exceeds a minimum threshold (≥ 0.4), preventing noise from low-quality inferences.

N8N: An open-source workflow automation platform enabling visual design of data pipelines. Used for orchestrating the detection→regulation→generation→verification process with transparent node-based architecture.

PostgreSQL: An open-source relational database system used for persistent storage of conversation sessions, personality states, and performance metrics, enabling longitudinal analysis.

Vector Embeddings: Numerical representations of text in high-dimensional space (typically 768-1536 dimensions for modern LLMs), enabling semantic similarity comparisons. Used by memory-based systems like mem0 and LangChain Memory.

mem0: An open-source memory layer for AI applications that stores user interactions as vector embeddings, enabling factual recall across sessions without explicit personality modeling.

LangChain Memory: A framework component for managing conversational context and history in LLM applications, supporting various memory strategies (buffer, summary, entity extraction).

Graceful Fallback: Error handling mechanism that ensures system failures produce neutral, supportive responses rather than crashes, maintaining user experience even during technical failures.

Document Statistics:

- **Total Length:** ~32 pages (estimated)
- **Word Count:** ~11,600 words
- **Sections:** 8 main sections plus comprehensive appendices (A-E)
- **Tables:** 9 numbered tables with captions and technical specifications
    - Table 1: Limitations vs. Contributions
    - Table 2: Evaluation Metrics and Targets
    - Tables 3-9: Technical specifications, workflows, risk matrix
    - Table B.1: Evaluation rubric (Appendix B)
- **Figures:** 2 diagrams with descriptive captions
    - Figure 1: System Architecture Pipeline (ASCII diagram)
    - Figure 2: N8N Production Workflow Architecture
- **Appendices:** 5 comprehensive appendices (A: Prompts, B: Rubrics, C: Technical Configs, D: Analysis Code, E: Glossary)
- **References:** 31 academic sources in APA 7th edition format
    - Coverage: 2010-2024 (emphasizing recent 2023-2024 LLM research)
    - DOIs provided for all journal articles and conference papers where available
    - Diverse sources: personality psychology, HCI, conversational AI, LLMs

- **Technical Depth:** Implementation-ready specifications with complete reproducibility documentation

This formal preliminary study document provides a comprehensive foundation for implementing and evaluating personality-aware conversational AI systems, with particular emphasis on transparency, reproducibility, and practical deployment in human-centered applications. All tables and figures are properly numbered and captioned for easy navigation and reference. Citations follow APA 7th edition guidelines with complete DOI links for verification and accessibility.